TECHNICAL SPECIFICATION

OF

SUPPLY OF 220V BATTERY SETS

SUPPLY OF 220V, 100AH VRLA BATTERY SETS

SUPPLY OF 220V, 200AH VRLA BATTERY SETS

TECHNICAL SPECIFICATION

(220V 100AH,200AH & 400 AH VRLA Battery sets)

1.0 SCOPE :

This specification covers design, manufacture, testing before dispatch, supply and delivery at site/CIP destination of Sealed Lead Acid (VRLA) Maintenance Free Battery sets as detailed below.

SUPPLY OF 220V, 100AH VRLA BATTERY SETS

SUPPLY OF 220V, 200AH VRLA BATTERY SETS

1.1 It is not the intent to specify completely herein all the details of the design and construction of material. However the material shall conform in all respects to high standards of engineering, design and workmanship and shall be capable of performing continuous commercial operation up to the expected life of the material, in a manner acceptable to the purchaser, who will interpret the meanings of drawings and specification and shall have the power to reject any work or material which, in his judgment is not in accordance therewith. The offered material shall be complete with all components necessary for their effective and trouble free operation. Such, components shall be deemed to be within the scope of Bidder's supply irrespective of whether those are specifically brought out in this specification and/or the commercial order or not.

2.0 **STANDARDS:**

2.1 The material shall conform in all respects to the relevant Indian Standard Specifications with latest amendments thereto.

Indian Standard	Title
IS 15549:2005	Stationery valve Regulated Lead Acid Batteries
ISS – 1885/1986 (Part-VIII)	Electro Technical vocabulary secondary cells and Batteries.
IEC60896-11:	Stationary Lead acid batteries
ISS – 1651/2013	Stationary Cells and Batteries Lead Acid type (with tubular (Fourth Revision)Positive plates.
ISS – 1069/2021	Water for storage Batteries
ISS – 8320/2000	General requirements and methods of test of lead acid storage Batteries (Latest version thereof)
IS-1652:2013	Lead Acid batteries with Plante' positive plates.
IS-266 : 1993	Sulphuric acid for storage batteries.
IS-1146:1981	Specification for rubber and plastics container for lead acid Storage batteries
IS:6071:1986	Synthetic separator for lead acid batteries

IEEE-485: 2020	IEEE Recommended practice for sizing of large lead acid storage batteries for generating stations and substations.
IEEE-484:	Recommended practice for design and installation of storage batteries.
IEC	International Electro Technical Commission Bureau Central De la commission electro Techique International. Rue de Cerembe, Geneva, Switzerland.
ISO	International Organisation for Standardization , Danish Board of Standardization, Aurehoegyej-12 DK-290, Hell Prup, Denmark.
ISS	Indian Standard Specification. Bureau of Indian Standards, Nanak Bhavan, 9, Bahadur Shah Zafar Marg, New Delhi – 100 002, India.

Equipment conforming to other internationally accepted standards, which ensure equal or higher quality than the standards mentioned above, would also be acceptable. In case the Bidders who wish to offer material conforming to the other standards, salient points of difference between the standards adopted and the specific standards shall be clearly brought out in relevant schedule. Four copies of such standards with authentic English Translations shall be furnished along with the offer. In case of conflict the order of precedence shall be (i) IS, (ii) IEC, (iii) Other standards. In case of any difference between provisions of these standards and provisions of this specification, the provisions contained in this specification shall prevail.

3. **SERVICE CONDITIONS:**

3.1 The material offered shall be suitable for operation under the following Service conditions.

i)	Location	:	At various locations in the State of Andhra Pradesh
ii)	Maximum ambient air temperature (deg C)	:	50
iii)	Minimum ambient air temperature of deg C	:	7.5
iv)	Average daily ambient Air temperature (deg.C0	:	35
v)	Maximum relative Humidity(%)	:	100
vi)	Maximum altitude above mean sea level (Metres)	:	1000
vii)	Average annual rain fall(mm)	:	925
viii)	Maximum wind pressure (kg/Sq.mm)	:	200
ix)	Isoceraumic level(days per year)	:	50

x) Seismic level(Horizontal accn.) : 0.10 g.

xi) Noise level : 45 db

The Batteries shall be for use in moderately hot and humid tropical climate, conducive to rust and fungus growth.

4. PRINCIPAL PARAMETERS:

4.1 The equipment shall conform to the following specific parameters:

Sl.No.	Item	Specification
1.	Type of Installation	In door
2.	DC voltage	220 V

5.0 **TECHNICAL REQUIREMNTS:**

The Batteries required under this specification are for supplying D.C. supply to control circuit, other D.C auxiliary power circuits in a EHV Sub-station. The batteries shall be capable of withstanding large discharge currents for operating various equipment. It shall be rugged construction designed for long life and for working satisfactorily without much alteration under the severest operating conditions and shall conform to the relevant ISS of latest issue.

The battery bank shall be supported in MS Modules/stand and supplied complete with all accessories.

5.1. TECHNICAL PARAMETERS:

The equipment shall conform to the following specific parameters:

S. No.	Description		220V- 100AH, 220V- 200AH and 220V
a)	No. of Cells per battery bank.	:	110 Nos. for VRLA
b)	Nominal voltage of each cell	:	2 volts.
c)	Nominal voltage of the complete bank	:	220 Volts
d)	Type of plates in each cell required.	:	VRLA flat pasted positive and pasted negative plates.
e)	Capacity of batteries (in Amp. hours at 10 Hour rate to an end voltage of 1.75V per cell for VRLA type).	:	100AH at 27 Deg. C- VRLA 200AH at 27 Deg. C- VRLA
f)	Cell dimensions and designation in accordance with the standard	:	Designation to be specified by the Bidder and maximum over all dimensions as per relevant standards. The bidder shall provide adequate thickness for the container to avoid any damages in transit & during installation particularly outer sides of the container where name plate and other details are pressed.
g)	Proposed method of working	:	i. Float charge (maximum of 235 Volt) and boost charge (with tap voltage of 200 Volts at 84th cell). ii. Float charge and quick charge (maximum load voltage should not exceed 235 Volts)
h)	Type of stands required and arrangement system for the batteries.	:	Metallic stands are required and Batteries are to be arranged in double row & double tier system.

VRLA (MAINTENANCE FREE) TYPE

DESIGN AND CONSTRUCTIONAL FEATURES OF BATTERY

Type : The battery shall be Valve Regulated Lead Acid Battery.

CONSTRUCTION

Battery

The battery shall be valve regulated sealed maintenance free lead acid type with lead cadmium antimony or lead calcium tin alloy for positive plate and lead calcium alloy for negative plate. Any other time proven alloy used for manufacture of positive and negative plate shall also be acceptable subject to providing the documentary proof. The battery shall be designed with excess negative active material truly emitting positive electrode to evolve oxygen first during charging. The negative electrode shall never become fully charged and evolve hydrogen. The sealed battery shall provide means to immobilize the electrolyte in such a way that the passage/voids remained present oxygen to diffuse to the negative plate. The battery offered shall be suitable for satisfactory operation at the ambient temperature of 50 deg. C. The rating of each individual cell offered shall be as per the specification at 10 hours discharge rate at an ECV of 1.75V. The battery shall be supplied in factory filled and charged condition. The bidder shall ensure that full battery capacity as per 5.1 of IEC: 896-2 shall be obtainable on first load cycle.

The supporting calculations for discharge characteristics of cell/battery for different time durations (10 Hrs, 3 Hrs, 2 Hrs. 1 Hr. & 1 minute) at an End cell voltage(ECV) of 1.75V/Cell shall be furnished with the bid.

The battery shall be complete with polypropylene co-polymer flame retardant container, lead plate, separators, vent plugs, inter-cell connectors, interbank connector bolts & nuts etc. as required.

a) Separators:

Separator shall be made of spun glass, microporous matrix and shall be resistant to sulphuric acid. It shall be capable of keeping all the electrolyte absorbed in the separator with no free electrolyte. It shall be electrically insulated. Internal resistance shall ensure discharge characteristics under all operating conditions. Sufficient separator overlap to edges of the plates is to be provided to prevent short circuit between the edges of adjacent plates.

b) Terminal Pillars:

The terminal pillars shall be suitable for mechanical connections with lead or lead alloy reinforced with copper/brass core inserts and current density shall not exceed 15A/mm2 at 1 minute rate.

c) Containers and lids

Container shall be made of polypropylene co-polymer (PPCP) material and shall have chemical and electrochemical compatibility and fire retardant. They shall be sufficiently robust and not liable to deformation under internal operating pressures and within the temperature range naturally encountered, leak proof, non-absorbent, resistant to the acid and free from flaws. They shall be able to withstand without fracture for 5 hours at 200 C at an

internal pressure of 5 times the normal operating pressure. The bidder shall indicate the pressure which the enclosure is capable to withstand in the bid. The list of tests and test certificates performed by the manufacturers of container shall be enclosed along with the bid.

The cell lids shall be made of same plastic material as that of container material and suitable either for resin bonding type or permanently sealing type based on the cell sealing design adopted by the manufacturer.

d) Pillar seal assembly

The pillar to lid seal assembly shall be designed to remain gas tight and electrolyte-tight during the design life of the units. When tested in accordance with relevant standards, the seal assembly shall after 200 days on test, show no signs of acid penetration.

e) Venting Device – Pressure regulating valve

The vent plug shall be made with suitable grade of polypropylene copolymer plastic material. The venting device shall be designed to relieve the excessive pressure from the cell. Seating of valve either shall be integral part of the cell lid or a provision in lid for separate venting device. Each cell shall be provided with venting device such that it cannot be opened without a proper tool. The valve shall be capable to withstand the internal pressure specified by the manufacturer.

Each valve regulated venting device shall be provided with flame arrestor capable of preventing the ingress of moisture / flame into the cell interior when the valve release the gas mixture..

The opening and closing pressure shall be within + or – 1 psi variation and manufacturer shall declare the closing and opening pressure values.

f) Type and thickness of positive plates:

Flat pasted plate type with high corrosion resistant lead cadmium antimony or lead calcium tin alloy or any other time proven alloy of suitable thickness to meet the AH capacity of the cell.

g) Type and thickness of negative plates:

The negative plates shall be of flat pasted type. The pasting shall be done on an automated machine for better control of process parameters. It should have adequate mechanical strength and would be so designed that active material is maintained in intimate contact with the grid under normal working conditions throughout the life of the battery. Lead calcium alloy or any other time proven alloy of suitable thickness to meet the AH capacity of the cell.

h) Electrolyte:

The electrolyte shall be battery grade sulphuric acid conforming to latest edition of relevant IS 266 and IS 1069

i) Battery enclosure:

The battery enclosure shall be designed to ensure efficient heat dissipation of air circulation. This shall be spelt out clearly in the bid.

j) Battery Racks (For VRLA)

The battery racks shall be constructed with high strength good quality mild steel sections. These battery racks shall be painted by the bidder with three coats of acid/alkali resistant

paint of approved make. The racks shall be of double row and double tier construction depending on the final layout based on space availability.

Connectors and Fasteners: The connections between the cells shall preferably be bolted type. The bolts and nuts, the connectors etc. shall be effectively lead-coated to prevent corrosion. Minimum thickness of lead coating shall be 25 microns in accordance with Appendix-F of IS 6848. Terminal plates of each cell shall be clearly stamped as positive and negative. All connectors and lugs shall be capable of continuously carrying the 30 minutes discharge current of battery and through short circuit current produced by the battery. The bidder shall furnish necessary calculation in support of this requirement.
Connectors may be supplied either as separate bars or alternately as part of the cell structure i.e. as suitably elongated heavy-duty copper strips. The connectors shall be lead coated copper of suitable size to join the cells. The connectors which shall be of suitably, designed copper connectors and suitably coated to withstand corrosion due to Sulphuric Acid and may be used where the cells are called upon to discharge at very high rates. The coating should be adequate and tenacious.

The following information shall be indelibly marked on outside of each cell:

a) Manufacturer's name, address and trade mark.
b) Country and month & year of manufacture
c) Serial number
d) Nominal voltage
e) AH capacity at 10 hr. discharge rate, ECV etc.
f) Polarity marking
g) Voltage of float operation

Enclosure/ Modules:
Batteries shall be installed in MS Enclosure/Modules to be supplied by the contractor to fit in the battery/battery charger room. Layout of batteries in the battery room shall be approved by the purchaser.

Enclosure /modules shall be so designed that cells are located within easy reach at convenient height.

Minimum number of bolts and nuts should be used primarily for anchoring and joining. Joining shall ensure proper and tight fit.

The supplier/Bidder shall provide insulator material for the bottom channel to avoid leakage path from battery to earth.

In case cell arrangement in the steel enclosure is horizontal, the steel enclosure shall be provided with flame retardant transparent front cover to prevent accidental contacts with the battery terminal.

Sealed batteries employing any new technology and meeting the performance requirement stipulated in the specification shall only be acceptable. However, the Bidder has to submit all relevant type test reports and copies of relevant standards along with the bid. The Bidder may furnish any additional technical particulars relevant to the technology of the battery offered in addition to the technical particulars as asked for in schedule of technical particulars in the bid.

5.2 ACCESSORIES AND SPARES REQUIRED

The following accessories shall be supplied along with each battery set shall be supplied, at free of cost.

i)	Suitable spanners for the Battery Terminals and bolts	2 Nos. per Set.
ii)	Inter Cell inter row, interbank connectors of adequate Size and mechanically strong of hard drawn copper Rods or strips, lead coated along with 25% of connectors and Bolts & Nuts as spare should to be provided as spare.	1 Set.
iii)	Rubber gloves(1100 V of reputed make)	2 Sets.
iv)	Battery O&M Manual	2 Nos.

5.3 The Battery shall be offered complete with all parts that are necessary or usual for their efficient operation. Such parts shall be deemed to be within the scope of this specification, whether specifically mentioned or not.

5.4 All parts particularly removable ones shall be inter changeable with each other.

5.5 The rating of the Batteries is specified at 27 deg. C. However, the battery shall be capable of operating satisfactorily (at reduced capacities) between 5 deg C and 50 deg. C and in locations where the relative humidity varies between 12% and 100%.

5.6 Auxiliary Power supply: The following auxiliary supply will be available at the substations where these batteries are to be located.

5.7 400 Volts (+ 10 or –15%), 3 phase, 4 wire 50C/S A.C with a solidly grounded neutral system for supply to the Battery Charger.

5.8 230 Volts (+ 10 or -15%), single phase, 50 C/S A.C. with solidly grounded neutral system for space heaters and A.C. control circuits.

5.9 The cells shall be suitably marked as per I.S.S.

5.10 VOLTAGE: The cell voltage shall not exceed 2.25 volts with a continuous low rate floating charge and shall not be less than 1.75 volts at the end of the emergency discharge.

5.11 OPERATION:

The D.C Battery shall be operated without an intentional ground. For indicating the incidence and degree of a ground fault on the D.C control circuitry, the mid-point of the battery shall be earthed through an ammeter of high resistance. The high resistance shall be so proportioned that the current flowing under the worst earth fault shall not exceed 250 mA.

6.0 TESTS:

6.1 All the offered equipment shall be fully type tested by the bidder as per the relevant standards including the type tests mentioned below. The type tests must have been conducted from NABL accredited laboratory within 7 years (as per CEA) prior to the date of bid opening. The bidder shall furnish two sets of type tests reports as per the relevant standards for each equipment offered along with bid. The offers received without these type tests reports shall be treated as non -responsive.

Type tests: As per IEC 60896-22.2004

The following shall constitute type tests:

S. No	Description of Test
1	Gas emission
2	High current Tolerance
3	Short circuit current and D.C internal resistance
4	Protection against internal ignition from external spark source
5	Protection against ground short propensity
6	Content & durability of required markings
7	Material identification
8	Valve operation
9	Flammability rating of materials
10	Intercell connector performance
11	Discharge capacity
12	Charge retention during storage
13	Float service with daily discharges for reliable mains power
14	Recharge behaviour
15	Service life at an operating temperature of 40° C for brief duration exposure time
16	Impact of stress temperature of 60° C for brief duration exposure time with 3 hr rate discharge test
17	Abusive over- discharge
18	Thermal runaway sensitivity
19	Low temperature sensitivity
20	Dimensional sensitivity at elevated internal pressure and temperature
21	Stability against mechanical abuse of units during installation.

Tests shall be conducted in accordance with IEC 60896-21 & IEC 60896-22,2004

List of Factory & site Tests for Battery

Sl. No.	Test	Factory Tests	Site Tests
1	Physical Verification		□
2	C/10 Capacity test on Battery bank	□	
3	C/10 Capacity test on Battery bank		□

Routine & Acceptance tests as per IS 1651:2013

i. Marking and Packing
ii. Verification of dimensions
iii. Test for capacity
iv. Test for voltage during discharge

Installation and commissioning

Manufacturer of Battery shall supervise the installation and commissioning and perform commissioning tests as recommended in O&M manual/ or relevant standards. All necessary instruments, material, tools and tackles required for installation, testing at site and commissioning are to be arranged by Battery manufacturer/contractor

7.0 INSPECTION:

7.1 The inspection may be carried out by the purchaser at any stage of manufacture. The successful Bidder shall grant free access to the purchaser's representative at a reasonable time when the work is in progress. Inspection and acceptance of any equipment under this specification by the purchaser, shall not relieve the Bidder of his obligation of furnishing equipment in accordance with the specification and shall not prevent subsequent rejection if the equipment is found to be defective.

7.2 The Bidder shall keep the purchaser informed in advance, about the manufacturing programme so that arrangement can be made for inspection.

7.3 The purchaser reserves the right to insist for witnessing the acceptance/routine testing of the bought out item.

7.4 The bidder shall give 15 days advance intimation to enable the purchaser to depute his representative for witnessing the acceptance and routine tests. The inspection charges would be to the suppliers account.

7.5. To & Fro Charges and accommodation shall be provided by the supplier at not lesser than APTransco TA rules.

7.6. APTRANSCO Engineer will not inspect incomplete materials, if APTRANSCO Engineer returned without inspections of offered equipment due to non-readiness of material at their works, the financial loss to the APTRANSCO for deputing Engineer representative will be recovered from the concerned supplier.

7.7 In case of any discrepancy, APTRANSCO has the right to re test the Finished Goods (Type Test and Acceptance Test) at the Cost of Bidder.

8.0 **QUALITY ASSURANCE PLAN:**

8.1 The bidder shall invariably furnish the following information along with his offer failing which the offer shall be liable for rejection. Information shall be separately given for individual type of equipment offered.

i) The Structure of organization

ii) The duties and responsibilities assigned to staff ensuring quality of work

iii) The system of purchasing, taking delivery and verification of materials

iv) The system for ensuring quality of workmanship

v) The quality assurance arrangements shall confirm to the relevant requirement of ISO-9001 on ISO 9002 as applicable.

vi) Statement giving list of important raw materials, names of sub-supplies for the raw materials, list of standards according to which the raw material are tested, list of tests normally carried out on raw material in the presence of suppliers representative, copies of test certificates.

vii) Information and copies of test certificates as on (i) above in respect of bought out items.

viii) List of manufacturing facilities available

ix) Level of automation achieved and list of areas where manual processing exists.

x) List of areas in manufacturing process, where stage inspections are normally carried out for quality control and details of such tests and inspections.

xi) List of testing equipment available with the bidder for final testing of equipment specified and test plant limitation, if any vis-a-vis the type. Special acceptance and routine tests specified in the relevant standards. These limitations shall be very clearly brought out in "Schedule of Deviations" from the specified test requirements.

8.2 The Supplier shall within 30 days of placement of order, submit the following information to the purchaser.

i) List of the raw material as well as bought out accessories and the names of sub-suppliers selected from those furnished along with the offer.

ii) Type test certificates of the raw material and bought out accessories if required by the purchaser.

iii) Quality Assurance Plan (QAP) withhold points for purchaser's inspection. QAP and purchasers hold points shall be discussed between the purchaser and contractor before the QAP is finalised.

The supplier shall submit the routine test certificates of bought out accessories and central excise asses for raw material at the time of routine testing if required by the purchaser and ensure that the quality assurance requirements of specification are followed by the sub-contractor.

8.3 The Quality Assurance Programme shall give a description of the Quality System and Quality Plans with the following details.

i. Quality System:

- The Structure of the Organisation
- The duties and responsibilities assigned to staff ensuring quality of work.
- The system of purchasing, taking delivery of verification of materials.
- The system of ensuring of quality workmanship.
- The system of control of documentation.
- The system of retention of records.
- The arrangement of contractor internal auditing.
- A list of Administrator and work procedures required to achieve contractors quality requirements. These procedures shall be made readily available to the purchaser for inspection on request.

ii) Quality Plans:

- An outline of the proposed work and program sequence.
- The structure of contractors organizations for the contract.
- The duties and responsibilities ensuring quality of work
- Hold and notification points.
- Submission of engineering documents required by this specification.
- The Inspection of the materials and components on request.
- Reference to contractors work procedures appropriate to each activity.
- Inspection during fabrication/ construction.
- Final inspection and test.

8.4 ~~Training of Personnel:~~

~~The contractor shall provide necessary facilities for training personnel at their works/principals works relating to testing, operation and maintenance for four personnel free of cost. However, travel and incidental charges of the personnel will be borne by the purchaser.~~

9.0 DOCUMENTATION:

9.1 All drawings shall conform to International Standards Organisation (ISO) 'A' series of drawing sheet/Indian Standards Specification. All drawings shall be in ink and suitable for micro filming. All dimensions and data shall be in S.I. Units.

9.2 List of Drawings and Documents:

The Bidder shall furnish one set of the following along with his bid.

i) General outline and assembly drawings.

ii) Graphics showing the performance of material.

iii) General constructional features.

iv) Type test reports

vi) Test reports literature, pamphlets of the bought out items & raw materials.

9.3 The Bidder submit four sets of final version of all the above drawings for Purchaser's approval. The Purchaser shall communicate his comments/approval on the drawings to the Supplier within two weeks of receipt of drawings. The supplier shall, if necessary, modify the drawings and resubmit four copies of the modified drawings for Purchaser's approval .The purchaser shall communicate approval with in two weeks of receipt of modified drawings. After receipt of Purchaser's approval, the Supplier shall, within three weeks, submit 4 prints of the approved drawings for Purchaser's use.

9.4 Six sets of the type test reports, duly approved by the purchaser, shall be submitted by the Bidder for distribution before commencement of supply. Adequate copies of acceptance and routine test certificates, duly approved by the purchaser, shall accompany the dispatch consignment.

9.5 The manufacturing of the equipment shall be strictly in accordance with the approved drawings and no deviation shall be permitted without the written approval of the Purchaser. All manufacturing and fabrication work in connection with the equipment prior to the approval of the drawing shall be at the Bidder's risk.

9.6 The Bidder shall also furnish four copies of bound manuals covering erection, commissioning, operation and maintenance instructions in English Language and all relevant information and drawings pertaining to the main equipment as well as auxiliary devices. Marked erection drawings shall identify the component parts of the equipment as shipped to enable Purchaser to carryout erection with his own personnel. Each manual shall also contain one set of all the approved drawings, type test reports as well as acceptance reports of the corresponding consignment dispatched.

9.7 Approval of drawings by the Purchaser shall not relieve the supplier of any of his responsibility and liability for ensuring correctness and correct interpretation of the drawings for meeting the requirements of the latest revision of the applicable standards, rules and codes of practices.

9.8 Raw Materials and bought out Items should be approved by APTRANSCO for utilization in manufacture of Batteries.

The equipment shall conform in all respects to high standards of engineering, design, workmanship and latest revisions of relevant standards at the time of supply and Purchaser shall have the power to reject any material which, in his judgment, is not in full accordance therewith.

10.0 PACKING, FORWARDING AND UNLOADING:

The equipment shall be packed in suitable crates so as to withstand handling during transit. The supplier shall be responsible for any damage to the equipment during transit, due to improper and inadequate packing and handling. The easily damageable materials shall be carefully packed and marked with the appropriate caution symbols. Wherever necessary, proper arrangement for lifting, such as lifting hooks etc., shall be provided. Any material found short inside the packing cases shall be supplied by the Supplier without any extra cost.

Each consignment shall be accompanied by a detailed packing list containing the following information:

a) Name of the consignee.
b) Details of consignment.
c) Destination.

d) Total weight of consignment.
e) handling and unpacking instructions.
f) Bill of material indicating contents of each package and spare material.

The Supplier shall ensure that the packing list and bill of material are approved by the Purchaser before despatch.

The Supplier shall arrange for unloading the material / equipment at his cost.

11.0 QUANTITY AND DELIVERY REQUIREMENTS:

i) This is set out in the proforma given in Price Schedule and the prices shall be quoted accordingly.
The scope of supply shall include supply of 25% extra-quantity of bolts, nuts, washers, split pins, cotter pins and such other small loose items free of cost.

12.0 SUPERVISION SERVICES:

The supplier shall provide services of qualified personnel for the supervision of erection, testing at site and commissioning the Battery Sets supplied duly conducting load Test and other relevant tests at site as per latest ISS.

13.0 Name Plate:

Equipment should be provided with name plate giving full details of manufacturer, capacities and other details as specified in the relevant ISS or other specification stipulated. The contract No. and date and year of supply and the words **"APTRANSCO"** must be etched on the name plate.

14.0 GUARANTEED TECHNICAL PARTICULARS:

The Guaranteed Technical particulars for the equipment being supplied shall be provided with the Bid as specified in this technical specification at Annexure-I**.** The Bids received without guaranteed technical particulars shall be treated as Non-responsive.

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A N N E X U R E – 1

GUARANTEED TECHNICAL PARTICULARS

FOR BATTERIES

SUPPLY OF 220V, 100AH VRLA BATTERY SETS
SUPPLY OF 220V, 200AH VRLA BATTERY SETS

S.No.	Description	100AH (VRLA)	200AH (VRLA)
1	Manufacturer's Name		
2	Applicable Standards		
3	Battery Voltage		
4	Battery Rating		
5	Designation as per ISS		
6	Manufacturer's type Design.		
7	A.H. Capacity at 10Hrs. Rate of discharge.		
8	No. of Positive plates/cell		
9	No. of Negative plates/Cell		
10	Total No. of Plates/Cell.		
11	Capacity in Ampere of battery Amps/end voltage a) 1 Minute load b) 1 Hour load c) 2 Hour load d) 3 Hour load e) 5 Hour load f) 8 Hour load g) 10 Hour load.		
12	Nominal cell voltage		
13	No. of Cells in each bank		
14	Internal resistance for each cell		
15	Resistance of the battery including with connection between the cells		
16	Short circuit current (Amps) Material of container		

Thickness, type and material
separator.

17 Electrolyte required for first
filling (Incl. 10% extra)

18 Special gravity of the electrolyte
at 27 Deg. C. with all cells fully charged

b) Special gravity of the electro-
Lyte at the end of discharge
at 10 Hrs. rate.

19 a) Construction details, dimension

a) Positive
b) Negative

b) Method of supporting elements

20 Surface area of plates, sq.mm

21 Wt. Of Electrolyte/cell in kgs.

22 Clearance between edges of
Plates and inner surface of
container.(incase of lead lined wooden container clearance between edges of plates and hangers mm)

23 clearance between bottom of
negative plates and bottom of
container mm.

24 Clearance between top of plates
and top of container mm.

25 Sediment space (depth) mm

26 Dimensions of each cell
a) Length mm
b) Width mm
c) Height mm
d) Thickness of container mm

27 Distance between centres of
cells where erected mm.

28 Nett Weight of the cell complete
with acid.

29 a) AH. Efficiency %
b) Wh. efficiency %
30 Recommended float charge

current and voltage Amp. Volts

31 Recommenced boost charge current and voltage Amp. Volts

32 Time required for boost charging from discharge condition

33 Maximum charging current/cell Amp.

34 Nominal charging rate Amp.

35 Maximum charging rate Amp.

36 Type of inter cell connection

37 Type of spray arrestors

38 Type of stand

39 Type of supporting insulators

40 Whether explosion proof or vent plugs provided.

41 Expected life span of battery

42 Rack details
a) No. of units
b) Description
c) Unit weight. Kgs.(shipping)
d) Unit length mm
e) Unit width mm
f) Unit height mm.

43 Ventilation required in battery room.

ACRONYMS

Reference Abbreviations	Names & Address
IEC	International Electro Technical Commission Bureau Central de la Commission electro Technique International 1. Rue de cerembe Geneva, Switzerland.
ISO	International Organisation for Standardization Danish Board of Standardization Aurehoegyej-12 Dk-290, Heel Prup DENMARK
ISS	Indian Standard Specification. Bureau of Indian Standards Nanak Bhavan, 9, Bahadur Shah Zafar Marg NEW DELHI – 100 002 INDIA
AC	Alternate current
DC	Direct current
V	Volt
VAC	AC Volts
A	Amps
Hz	Hertz
mV	milli Volts
M.Ohms	Mega Ohms

SECTION - V

TECHNICAL SPECIFICATION

OF

SUPPLY OF 220V BATTERY CHARGERS

LOT-I: SUPPLY OF 220V, 100AH (10A FC / 20A FCBC) BATTERY CHARGERS with separate DCDB Panel & dropper diode in separate enclosure

LOT-II: SUPPLY OF 220V, 200AH (25A FC / 50A FCBC) BATTERY CHARGERS with separate DCDB Panels & dropper diode in separate enclosure

SECTION-V

TECHNICAL SPECIFICATION

For 220V, 100AH Battery charging Equipment and separate DC Distribution Board

1.0 SCOPE:

This specification covers design, engineering, manufacture, stage testing, inspection and testing before supply and delivery at site/CIP destination of following Battery charging equipment suitable for the Maintenance Free Valve Regulated Lead Acid Battery sets and Tubular type Battery sets **(VRLA/TUBULAR)**.

220 V 100AH Battery charging equipment comprising separate Float Cum Boost Charger & separate Float Charger suitable for 220 Volts 100AH Maintenance Free Valve Regulated Lead Acid type batteries and Tubular type Battery sets with separate DC Distribution boards & dropper diode in separate enclosure

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2.0 STANDARDS:

2.1 The material shall conform in all respects to the relevant Indian Standard Specifications with latest amendments thereto.

Indian Standard	Title
IS 16539/2017 (Part1)	Semiconductor Converters.
IS 6619	Safety Code for Semiconductor Rectifier Equipment
IS/IEC 60947	Low-voltage switch gear and control gear.
IS/IEC 61558	Safety of power transformers, power supply units and similar
IS/IEC 60269	Low-Voltage Fuses for Voltages not Exceeding 1000 V AC or 1500 V DC

Equipment conforming to other internationally accepted standards, which ensure equal or higher quality than the standards mentioned above, would also be acceptable. In case the Bidders who wish to offer material conforming to the other standards, salient points of difference between the standards adopted and the specific standards shall be clearly brought out in relevant schedule. Four copies of such standards with authentic English Translations shall be furnished along with the offer. In case of conflict the order of precedence shall be (i) IS, (ii) IEC, (iii) Other standards. In case of any difference between provisions of these standards and provisions of this specification, the provisions contained in this specification shall prevail.

3. CLIMATIC CONDITIONS:

The material offered shall be suitable for operation under the following Service conditions.

i)	Location	:	At various locations in the State of Andhra Pradesh
ii)	Maximum ambient air temperature (deg C)	:	50
iii)	Minimum ambient air temperature (deg C)	:	5
iv)	Average daily ambient Air temperature (deg.C)	:	35
v)	Maximum relative Humidity(%)	:	80
vi)	Maximum altitude above mean sea level (Metres)	:	1000
vii)	Average annual rain fall(mm)	:	940
viii)	Maximum wind pressure (kg/Sq.mm)	:	200
ix)	Isoceraumic level(days per year)	:	50
x)	Seismic level(Horizontal accn.)	:	0.10 g.
xi)	Noise level	:	45 db

The Battery Chargers shall be for use in moderately hot and humid tropical climate, conducive to rust and fungus growth.

4. PRINCIPAL PARAMETERS:

The equipment shall conform to the following specific parameters:

Sl.No.	Item	Specification

1.	Type of Installation	In door
2.	AC voltage	415 V +20% to –20% (for 220 V batteries)
3.	No. of phases	Three
4.	Frequency	50 c/s ± 5%
5.	Regulation	± 1%
6.	Ripple content	2% RMS without Battery connected

5.0 TECHNICAL REQUIREMNTS:

5.1 The Battery charging equipment shall comprise of a Float Charger and a Float Cum Boost Charger.

General remarks:

1. **The following are the parameters to be displayed in Multi-Function digital meters**

   AC Voltage parameters: AC input voltage and currents (All three phases), power and frequency-1No. may be displayed in digital multi-function meter. All AC power supply wires should have colour coding i.e. Red, Yellow, Blue, and Black.

   DC Voltage parameters: DC Voltage and currents (FC, FCBC, Load, Battery)-4 Nos. 2 Line display meters individually for FC, FCBC, Load and Battery. All DC power supply wires should have colour coding i.e. Red for Positive and Black for Negative.

   The above parameters are to be communicated to SAS.

2. **The following are the parameters to be displayed in Analog meters**

   Earth Leakage current, Battery current (Indicating whether charging/ discharging).

   The Earth leakage current is to be communicated with SCADA by providing a Transducer with 4-20 milli amp output.

3. **All the alarm, trip and status signals are to be communicated to SAS. An annunciator panel shall be provided for this purpose.**

4. **For setting of Float/boost voltages, separate potentiometers are to be provided for both VRLA and Tubular type batteries i.e. one for setting float voltage in FC mode for VRLA, one for setting float voltage in FC mode for Tubular, one for setting float voltage in FCBC mode for VRLA, one for setting float voltage in FCBC mode for Tubular, one for setting boost voltage in FCBC mode for VRLA, one for setting boost voltage in FCBC mode for Tubular.**

5. **Short circuit protection shall be provided. The charger shall withstand during short circuit condition. The same shall be demonstrated at the site during commissioning along with other logics and protections and same shall be recorded**

in MOM. The charger shall either limit the current or turn off the gate pulses of the charger during short circuit condition.

6. **Kick fuses /Spring-loaded fuses are not allowed.**
7. **Terminal blocks shall be of ELMEX-KABTM4 or M/s Connect well STH4 .**
8. **All internal wiring to be routed through cable trunk.**
9. **Provision for accept/reset of annunciation from Remote maybe arranged. (or) Potential Free Contacts shall be provided for Alarm Accept & Alarm Reset.**
10. **Provide rectifier fuse with base and carrier.**
11. **Provide internal General arrangement diagram.**
12. **The Positive and negative DC busbars maybe separated from each other by approx. one foot.**
13. **For Earth fault protection, a separate relay shall be used which can be replaced easily without disturbing any other circuits .**
14. **DCDB should be supplied separately.**
15. **3 Pole Battery isolator switch/MCCB is to be provided in Battery circuit which shall be suitable for withstanding 30 minutes battery discharge current rate and also MCB/MCCB of battery output rating not less than 63 Amps for 100AH, 100 Amps for 200AH battery sets.**

    The power circuit, MCBs, cables etc between battery terminals to the load including the cable connecting charger to DCDB and also the internal ckts of DCDB shall be adequately designed to meet 30 minutes of battery discharge current and through short circuit current produced by the battery bank, and other circuits shall be as per the charger rating.
16. **All internal components are to be mounted on channels for easy replacement. All internal power circuits (DC/AC) shall be adequately designed to meet 30 minutes of Battery discharge current and through short circuit current produced by the battery bank.**
17. **The selection of modes like AUTO/MANUAL, VRLA/Tubular, FC/FCBC shall be done through switch only. Auto / Manual selection and VRLA / NVRLA selection switches (3 No.) shall have a lock & key facility to avoid mal-operation.**
18. **Battery current limiter shall be provided and set as per C10 rating of battery set and also having the provision for adjusting the battery current limiter setting.**
19. **There shall not be any Auto Mode in Charger for NVRLA/Tubular Battery Units. If the operator unknowingly select this mode, the Charger shall operate in Manual Mode only. After selecting NVRLA battery mode the control logics of VRLA mode shall be isolated from charger circuit.**

    The following are the details of Float / Boost voltages for Batteries.

	VRLA type	**Tubular type**
Float voltage set point (per cell)	**2.05 V to 2.3 V.**	**2.05V to 2.3 V.**

Boost Voltage set point(per cell)	**2.15 V to 2.4 V.**	**2.15V to 2.7 V.**
Float voltage range	**190 V to 250 V**	**190 V to 250 V**
Boost voltage range	**240 V to 265 V**	**240 V to 305 V**

5.1.1 FLOAT CHARGER:-

The Float Charger is to operate on three phase A.C. supply of 415 Volts with +20% to –20% variation, 50 C/S + 5%. The Float Charger shall be capable of Floating the Battery i.e., trickle charge the Battery comprising of 110Nos., 2Volts, 100AH lead acid cells at about 2.23v per cell and simultaneously supplying a constant load of 10 Amperes in the climatic conditions specified in the above clause(3) . The Charger shall be provided with a regulator to facilitate controlling of the cell voltage and to stabilize the output voltage within + or - 1% of the set D.C. value, for A.C mains supply voltage variation of +20% to –20% and frequency variation of + or - 5% and D.C. load variation of 0-100% and also when all the three variations occur Simultaneously.

The Float Charger shall be provided with an automatic current limiting facility such that when Float Charger output current exceeds 10% above the rated current, the Float Charger voltage should be brought down automatically so that the Float Charger output current does not exceed the set value. The Float Charger shall be of 3 phase full wave full-controlled thyristor bridge rectifier type with automatic voltage regulator unit and with necessary printed circuit boards, transformers and relays etc.

5.1.2 FLOAT CUM BOOST CHARGER: -

The Float Cum Boost Charger is to operate on a three phase supply of 415 Volts with + 20% -20% variation 50 C/S. The Float Cum Boost Charger shall be capable of Boost charging the battery comprising of 110 Nos. 2V 100AH lead acid cells upto a maximum cell voltage of 2.7v per cell at a maximum charging current of 20 Amperes for the Chargers suitable for 100AH batteries in the climatic conditions specified in the above clause(3). The Float Cum Boost Charger shall be capable of Boost charging the battery comprising of 110Nos., 2Volts, 100AH lead acid cells from fully discharged condition to fully charged condition as per C10 rating of battery set..

5.2 **The Battery charging equipment shall be complete with all parts that are necessary for their efficient operation. Such parts shall be deemed to be within the scope of this specification whether specifically mentioned or not.**

5.3 OPERATION

- A Charger should have 2 No independent charging circuits such as one for FC and one for FCBC.
- The charger should have AC over voltage protection relay with alarm and trip with time setting facility.
- Each charging circuit should be **3-ph full wave full controlled** bridge rectifiers (6 thysrisors) for smooth DC wave form

- Both charging circuits should be of constant voltage type with soft start and should have short circuit protection with current limiters.
- Each charging circuit should maintain constant DC set voltage (± 1%) with AC voltage regulation (+20% to -20%)
- Each charging circuit should maintain constant DC set voltage (± 1%) up to rated current limit set and above which the voltage should fall accordingly (drooping characteristic).
- 1st charging circuit named as **Float Charger (FC)** should always be ON and support the DC load connected to Charger along with trickle charging the battery.
- 2nd charging circuit named as **Float Cum Boost Charger(FCBC)** should be generally in OFF condition, up on raise of battery current drawn above a specified value the FCBC circuit should ON automatically by detaching battery from Float charging circuit through a DC contactor and should charge the batteries in high rate mode i,e BOOST mode. And up on full charging of batteries with the decrease of battery current drawl below the specified value FCBC should switch OFF with battery again connecting to Float Charger through DC contactor.
- In case of failure of FC, the FCBC should ON and function as Float Charger supplying DC loads with trickle charging the battery.
- FCBC should work as Boost Charger in case of main FC healthy and should act as Float Charger in case of FC failure thus fulfilling the name **Float Cum Boost Charger (FCBC)**.
- Charger should have **Auto/Manual Float/Manual Boost** mode of operation in VRLA type and charger should have **Manual Float/Manual Boost** mode of operation only for Tubular Type and separate mode selector switches shall be provided for VRLA type and Tubular type each and Battery type selector switch shall be provided for selecting VRLA Type or Tubular Type.
- In Auto mode (For VRLA only)
    - FC should switch ON automatically with AC supply ON
    - FCBC should ON automatically, for BOOST charging the battery, with increase of battery current and should OFF with decrease of battery current provided FC healthy only.
    - FCBC should ON automatically in case of failure of FC and should work as Float Charger and should OFF upon FC becomes healthy.
- In Manual Float mode(for VRLA type)
    - FC should remote OFF and FCBC should ON and function as Float charger for float charging the Battery and DC Contactor ON for Manual float selection of VRLA mode.
    - When FCBC OFF/Trip, FC should ON and function as FC charger.
- In Manual Boost mode (for VRLA)

- If FC is healthy and supporting DC loads, the FCBC should ON and function as Boost Charger, for BOOST charging the battery and DC Contactor OFF.
- If FC is FAIL/OFF/Trip, the FCBC should only work as Float charger and DC contactor ON.

- In Manual Float mode(for Tubular)
  - If FC is healthy and ON for supporting DC loads, the FCBC should be ON and function as float Charger,for float charging the Battery and DC Contactor OFF.
  - If FC is FAIL/OFF/Trip, the FCBC should only work as Float charger and DC Contactor ON.

In Manual Boost mode (for Tubular)

- If FC is healthy and ON for supporting DC loads, the FCBC should ON and function as Boost Charger, for Boost charging the Battery and DC Contactor OFF.
- If FC is FAIL/OFF/Trip, the FCBC should only work as Float charger and DC Contactor ON

- The DC contactor connecting battery to Float Charger/Loads should OFF only while **FC is healthy** & **FCBC is ON** either in Auto mode or Manual float/Manual Boost mode. The DC contactor shall follow as per the logic table given below.

- While FC feeding loads and FCBC is Boost charging the battery exclusively with DC contactor open, upon immediate failure of Ac supply/FC fail, the DC contactor should close immediately for delivering full DC voltage.
- The following logic matrix should be followed for the operation of charger.
- **The Manufacturer should submit the Logic (shall explain how they are providing all the protections as per APTRANSCO Specification) of all Protections available for the Battery Charger at the time of Applying for Drawing Approvals. The Approved Logic Drawing shall be Stickered, Painted on inside of the Front Door.**

LOGIC MATRIX FOR FC AND FCBC OPERATIONS (For VRLA mode)

System condition	Mode selection	Condition of FC	Condition of FCBC	Battery current (Set value)	Interlocking Contractor (K1)	Load	Battery
AC Supply Available	AUTO	Healthy	Healthy	Less than set value	ON	Supplied by FC (Float voltage)	On Float Voltage (By FC) and FCBC shall be Automatically OFF
AC Supply Available	AUTO	Healthy	Healthy	Greater than set value	OFF	Supplied by FC (Float voltage)	On Boost Voltage (By FCBC)
AC Supply Available	AUTO	FAIL	Healthy	Less/ Greater than set value	ON	Supplied by FCBC (Float voltage)	On Float Voltage (By FCBC)
AC Supply Available	AUTO	Healthy	FAIL	Less/ Greater than set value	ON	Supplied by FC (Float voltage)	On Float Voltage (By FC)

AC Supply Available	MANUAL FLOAT	Healthy	Healthy	Less/ Greater than set value	ON	Supplied by FCBC (Float voltage) and FC shall be Automatically OFF	On Float Voltage (By FCBC)
AC Supply Available	MANUAL BOOST	Healthy	Healthy	Less/ Greater than set value	OFF	Supplied by FC	On Boost Voltage (By FCBC)
AC Supply Available	MANUAL FLOAT/ BOOST	FAIL	Healthy	Less/ Greater than set value	ON	Supplied by FCBC (Float voltage)	On Float Voltage (By FCBC)
AC Supply Available	MANUAL FLOAT/ BOOST	Healthy	FAIL	Less/ Greater than set value	ON	Supplied by FC (Float voltage)	On Float Voltage (By FC)
AC Supply Available	ANY MODE	FAIL	FAIL	Less/ Greater than set value	ON	Supplied by Battery	On Discharge
AC Supply Fail	ANY MODE	Healthy	Healthy	Less/ Greater than set value	ON	Supplied by Battery	On Discharge

LOGIC MATRIX FOR FC AND FCBC OPERATIONS (For TUBULAR mode)

System condition	Mode selection	Condition of FC	Condition of FCBC	Interlocking Contractor (K1)	Load	Battery
AC Mains Available	MANUAL FLOAT	Healthy	Healthy	OFF	Supplied by FC (Float voltage)	On Float voltage (By FCBC)
AC Mains Available	MANUAL FLOAT	Healthy	FAIL	ON	Supplied by FC (Float voltage)	On Float Voltage (By FC)
AC Mains Available	MANUAL FLOAT	FAIL	Healthy	ON	Supplied by FCBC(Float voltage)	On Float voltage (By FCBC)
AC Mains Available	MANUAL BOOST	Healthy	Healthy	OFF	Supplied by FC (Float voltage)	On Boost Voltage (By FCBC)
AC Mains Available	MANUAL BOOST	Healthy	FAIL	ON	Supplied by FC (Float voltage)	On Float Voltage (By FC)
AC Mains Available	MANUAL BOOST	FAIL	Healthy	ON	Supplied by FCBC (Float voltage)	On Float Voltage (By FCBC)
AC Mains Available	ANY MODE	FAIL	FAIL	ON	Supplied by Battery	On Discharge
AC supply Fail	ANY MODE	Healthy	Healthy	ON	Supplied by Battery	On Discharge

5.4 The Float Charger for 220 Volts,100AH batteries shall comprise of the following components but not limited to them.

i) AC 3-pole Moulded Case Circuit Breaker of suitable capacity : 1 No.

ii) Pilot (Clustered LEDs) lamps to indicate : 3 Nos.

A.C. mains supply (Common for both FC&FCBC)

iii) 3 phase full wave Full-controlled thyristor bridge rectifier with fuses : 1 No.

iv) Automatic voltage regulator unit with necessary printed circuit boards transformers and relays, etc. : 1 Set

v) Filter choke and filter condenser : 1 Set

vi) Double wound impregnated natural air cooled three phase mains transformer of suitable rating : 1 No.

vii) D.C. 2 pole Moulded Case Circuit Breaker of suitable capacity : 1 No.

viii) A.C Power contactor and thermal overload relay - Each one number

Though not specifically included in the above list, the supplier of the battery Chargers shall supply all the required control equipment for trouble free operation of the Chargers..

5.5 The Boost Charger for 220 volts, 100AH batteries shall comprise of the following components but not limited to them.

i) AC 3-pole Moulded Case Circuit Breaker of suitable capacity : 1 No.

ii) Pilot (Clustered LEDs) lamps to indicate A.C. mains supply : 3 Nos. (Common for both FC&FCBC)

iii) 3 phase full wave Full-controlled thyristor bridge rectifier with fuses : 1 No.

iv) Automatic voltage regulator unit with necessary printed circuit boards transformers and relays, etc. : 1 Set

v) Filter choke and filter condenser : 1 Set

vi) Double wound impregnated natural air cooled three phase mains transformer of suitable rating : 1 No.

vii) D.C. 2 pole Molded Case Circuit Breaker : 1 No.

of suitable capacity

viii) A.C Power contactor and thermal overload relay - Each one number

5.6 The battery Chargers shall also comprise the following equipment.

i) 1 No. silicon Tap Cell diode connected to 107th cell for VRLA Battery and 93rd Cell for Tubular Battery and the D.C. positive bus to maintain continuity of D.C. supply to the D.C bus in the event of A.C. failure while Boost Charger in service and to avoid interruption short time of to connect 1st cell and the 107th cell for VRLA Battery and 93rd cell for Tubular Battery positive.

ii) Ripple filtering devices.

iii) 1 No. triple pole Molded Case Circuit Breaker of suitable capacity for connecting to 110th cell positive and 1st cell negative and 107th cell for VRLA Battery and 93rd cell for Tubular

iv) Internal light operated on 230 Volts single phase, 50 C/S, A.C. system with door opening/closing ON/OFF switch.

v) 2 Nos. earthling lugs suitable for receiving 120 sq.mm copper conductors.

vi) Space heaters suitable for operation AC 230 V, 1 Amp, 50 c/s.

vii) Heater ON/OFF switch.

viii) Undrilled Gland plate for all the external cables.

5.7 The Chargers shall have built-in automatic voltage control and load limiting features. The voltage regulator shall automatically sense, monitor and regulate the D.C. voltage to within + 1% of the set value from No load to full load and under A.C. input supply voltage and frequency fluctuations. Load limiting features shall automatically reduce the output voltage of the Charger on loads more than the rated load.

5.8 The ripple content of the Charger D.C. output shall not exceed 2% when the battery is not connected.

5.9 Alarm and fault indicating lamps shall be arranged in the following faults. **(**Facia Display Unit) with RS-485 port will be provided

a) Main incoming AC supply "ON"

b) DC supply "ON"

c) A.C. input Circuit Breaker fail both Float and Float Cum Boost Charger.

d) D.C. output Circuit Breaker fail both Float and Float Cum Boost Charger

e) Mains failure

f) Phase failure

g) D.C. output over voltage

h) D.C Under voltage

i) Rectifier fuse failure for both Float Charger and Float Cum Boost Charger

j) Filter condenser fuse failure for both Float Charger and Float Cum Boost Charger

k) Battery earth fault.

l) Battery MCCB Fail

m) DC contactor OFF

n) AC over Voltage Alarm

o) AC Under Voltage Alarm

Battery current Limiter:

The charger should have battery current limiter to limit the Battery charging current upto C10 Charging current with Battery current limit setting provision.

Load Limiters:

All chargers shall have load limiters which shall cause, when the voltage control is in automatic mode, a gradual lowering of the output voltage when the DC load current exceeds the load limiter setting of the Charger. The Load-limiter characteristics shall be such that any sustained overload or short circuit in DC System shall not damage the Charger.

Soft Start:

The charger shall be provided with soft start feature so as to make DC output voltage to rise gradually to its rated value over duration of 6-8 seconds whenever the charger is powered or at the time of restoration from shutdown. Soft start feature should be invariably provided to minimize the inrush current from the float charger to discharged battery.

Battery over voltage protection

Over voltage shutdown/cutback: **The charger should compare DC output voltage with over voltage set point, above which the charger should inhibit the firing pulse to charger thyristors so as to shut down the system thus avoiding over voltage at charger output. The same can be reset through over voltage reset in set parameters.**

Filter Circuit

Suitable filter circuits shall be provided in the chargers. The ripple content (peak to peak) in the output voltage shall not exceed 2%, irrespective of the DC load.

Contactors

All battery chargers shall be provided with AC contactors on the input side, shall be of air break type and suitable for continuous duty. The operating coil shall be rated for 415 volts +/- 20% AC.

AC Contactor shall be provided after the secondary of the rectifier transformer. Thermal overload relay shall control the contactor.

Thermal Overload Relay

A thermal overload relay incorporating a distinct single phasing protection (using differential movement of bimetal strips) shall also be provided for the AC input. The relay shall trip the above contactor

Rectifier Transformer

The rectifier transformer shall be copper wound Vacuum Impregnated continuously rated and natural air cooled (N.A.) type. The rating of the rectifier, transformer shall correspond to the rating of the associated rectifier assembly. The Bidder shall provide the supporting calculations for the transformer capacity during design & detailed engineering. Temperature rise of the transformer during fully loaded condition shall not cause damage to the electronic components of the charger. Necessary ventilation arrangements have to be made to limit the temperature rise to safe limits. There shall be separate transformers for FC charger and FCBC charger, physically located at the bottom of the panel for easy inspection.

Rectifier Assembly

Phase rectifier bridge assembly shall be full wave full controlled (6 thyristors) shall be provided for both float and float cum boost charger. The rectifier assembly shall be designed to meet the duty as required by the respective Charger. The rectifier shall be provided with their own heat sink having their own dissipation arrangements with natural air-cooling. Necessary surge protection devices and semi-conductor fuses shall be provided in each arm of the rectifier connections. It should be possible to easily replace individual thyristors. Each thyristor shall be fixed with bolted arrangement for easy testing and replacement.

Diode dropper circuit shall be provided in separate enclosure.

POTENTIAL FREE CONTACTS

The following Potential Free Contacts shall be available on FC & FCBC:

i) AC Mains Fail

ii) FC AC I/P Circuit Breaker Trip

iii) FCBC AC I/P Circuit Breaker Trip

iv) FC DC O/P Circuit Breaker Trip

v) FCBC DC O/P Circuit Breaker Trip

vi) Battery Earth Fault

vii) FC Rectifier Fuse Fail

viii) FCBC Rectifier Fuse Fail

ix) FC Filter Fuse Fail

x) FCBC Filter Fuse Fail

xi) DC Over Voltage

xii) DC Under Voltage

xiii) Battery Input Circuit Breaker Trip/Off

xiv) Group alarm

xv) Alarm Accept

xvi) Alarm Rest

xvii) DC Contactor OFF

xviii) Phase Fail

xix) AC over voltage Trip

xx) AC under voltage Trip

5.10 Necessary PCBs/relays for the above shall be provided wherever necessary.

5.11 A block diagram to indicate the general protection features provided for the Charger shall be furnished along with the Bid.

5.12 Necessary relays for annunciation for faults mentioned in 5.9 shall be provided. They shall be suitable for operation of 220 V D.C. supply with + 15% to –15% variation. Necessary circuitry for cancelling alarm shall be provided. On cancelling the alarm, the sound only should go and lamp should continue till the fault is rectified/cleared.

5.13 A Blocking diode of adequate rating shall be provided in the positive pole of the output circuit of each charger to prevent current flow from the DC Battery into the charger.

5.14 In the event of A.C supply failure, when Battery is on Boost charge, the available battery capacity shall be automatically connected to the D.C Bus

5.15 All fuses shall be HRC non-deteriorating type wherever necessary. Unless otherwise specified, the fuses shall be of class-4 (80KA prospective breaking current) for A.C. circuits and class-2(33 KA prospective D.C. current) for D.C. circuits.

MCB/MCCBs

i. AC input and DC output MCB/MCCBs shall be provided to in all chargers in addition to contactor and thermal over load relay and they shall be of heavy duty single through group operated quick make, quick break type complying with IS: 8828 & IEC: 898-1987 and suitable for 400/240 V A.C and 220 V DC. All MCCBs shall be suitable for continuous duty. MCCBs should have auxiliary contacts for annunciation.

ii. MCB/MCCBs shall be so designed with suitable breaking capacity in conformity with above IEC & ISI. The bidder shall ensure that all MCCB/MCBs are adequately rated so as to be fully protected during all abnormal operating conditions.

iii. **MCCB/MCB shall be flush mounted on Battery Chargers Panels.**

iv. Suitable door inter locking lamps shall be provided with micro switches.

v. MCCB shall be provided with bimetallic & magnetic type release for over current & short circuit protection. The setting of the thermal release shall be adjustable between 50% to 100% of the rated current.

vi) MCBs for A.C and D.C control supply and heaters wherever required shall be mounted inside the chargers.

5.16 The indicating instruments shall be of flush mounting type with dust tight covers. The instruments shall be approximately 96 mm square with 240 deg. C scale or alternatively 144 mm square with 90deg.C scale. Dials shall be white with black numbers and lettering.

5.17 **Transducers:** The contractor shall supply the Transducers for the measurement of Earth Leakage Current. The Output (4-20mA) of the transducers shall be carried up to Terminal Blocks for taking them to Auxiliary BCU for Automation.

5.18 The internal wiring of the Charger shall be carried out with PVC insulated 1100 V FRLSH grade stranded copper conductor. The control wiring shall be carried out with 2.5 sq.mm copper conductors. All wiring for external connections shall be brought to 1100 V grade single piece mounted terminal blocks with barriers. Suitable vertical terminal blocks shall be provided. Terminal connectors provided for wiring shall be stud type and not screw type.

5.19.1 All the external cables to be connected to the Charger shall be arranged for bottom entry.

The Float charging equipment and Boost charging equipment shall be mounted in a metal enclosed, sheet steel cubicle, indoor floor mounting, free standing type. It shall

be totally enclosed, completely dust tight, weather and vermin proof. The cubicle shall be adequately ventilated with louvers to facilitate the cooling of transformers and rectifiers. The ventilating openings shall be less than 3 mm size. The panel shall be complete with internal wiring, terminal board for internal and external connections. The panel shall be made out of suitable angles and M.S. Sheet of 16 SWG for the front,18 SWG for the back, top and sides, and 14 SWG for the bottom. The height of the panel shall be 1800 mm and depth 600 mm and panel shall have suitable width to accommodate the charging equipment. i.e., adequate clearance / space should be provided between item to item in the Charger, to facilitate easy handling at the time of attending on repairs. The Charger shall be provided with hinged type rear doors. Ingress protection of Charger shall meet with IP 42.

Each Charger shall be furnished completely wired upto power cable lugs and terminal blocks with insulated barrier between each terminals of AC Input, DC output and Battery input and ready for external connections. The control wiring shall be carried out with 1.1 KV PVC insulated, 2.5 sq.mm. Stranded copper wires, Control terminals shall be suitable for connecting two wires, with 2.5 sq.mm stranded copper conductors. All terminals shall be numbered for ease of connections and identification. Each wire shall bear a ferrule or tag on each end for identification. At least 20% spare terminals shall be provided for control circuits.

The insulation of all circuits except the low voltage electronic circuits shall withstand test voltage of 2 kV AC for one minute. An air clearance of at least ten (10) mm shall be maintained throughout for such circuits, right up to the terminal lugs. Whenever this clearance is not available, the live parts shall be insulated or shrouded.

The low voltage electronic circuits tests shall be as per relevant standards and the bidder has to furnish supporting documents for low voltage electronic circuits.

The battery Chargers shall be tested for its ratings before dispatch.

5.20. ACCESSORIES AND SPARES REQUIRED

The following accessories shall be supplied along with each battery Charger.

i) Suitable wrenches : 2 Nos. per Set.

ii) HRC Fuses, Glass Fuses, MCCBs, Thyristors and such small parts shall be provided 25% extra as spare.

iii) Battery Charger O&M Manual along with drawings : 2 Nos.

5.21. The Battery Charger shall be offered complete with all parts that are necessary or usual for their efficient operation. Such parts shall be deemed to be within the scope of this specification, whether specifically mentioned or not.

5.22 **PAINTING**: All cubicles shall undergo tank cleaning process according to latest IS in vogue followed by powdered coating process.

2.0 TESTS :

6.1 **Type Tests:**

6.1.1 All the offered equipment shall be fully type tested by the bidder as per the relevant standards including the type tests mentioned below. The type tests must have been conducted on the similar or higher capacity from recognized test laboratory within 7 years prior to the date of bid opening. The bidder shall furnish two sets of type tests reports as per the relevant standards for each equipment offered along with bid. The offers received without these type tests reports shall be treated as non -responsive.

a) Voltage Regulation Test
b) Load limiter characteristics Test
c) Efficiency Tests
d) High Voltage Tests
e) Temperature rise Test
f) Short circuit Test at no load and full load at rated voltage for sustained short circuit
g) Degree of protection Test
h) Measurement of ripple by oscilloscope

Additionally, If the type tests submitted are of different rating from the offered rating, then, all the above tests except the Ingress Protection test shall be carried in-house at factory premises on one charger in the presence of APTransco

6.1.2 Routine Tests: Each equipment manufactured shall be tested for (i) Insulation Resistance Test (ii) High Voltage Test (iii) Annunciation Verification (iv) DC output voltage regulation (v) Efficiency at full load (vi) Power factor (vii) DC output voltage ripple (viii) Battery path current limit and (ix) Battery Low Trip Setting (x) Protection against short circuit (xi) Temperature rise test

The supplier should produce all the relevant original IEC/IS copies of latest amendments/revisions purchased by them for reference by AP Transco representative during acceptance tests.

7.0 INSPECTION:

7.1 The inspection may be carried out by the purchaser at any stage of manufacture. The successful Bidder shall grant free access to the purchaser's representative at a reasonable time when the work is in progress. Inspection and acceptance of any equipment under this specification by the purchaser, shall not relieve the Bidder of his obligation of furnishing equipment in accordance with the specification and shall not prevent subsequent rejection if the equipment is found to be defective.

7.2 The Bidder shall keep the purchaser informed in advance, about the manufacturing programme so that arrangement can be made for inspection.

7.3 The purchaser reserves the right to insist for witnessing the acceptance/routine testing of the bought out item.

7.4 The bidder shall give 30 days advance intimation to enable the purchaser to depute his representative for witnessing the acceptance and routine tests. The inspection charges would be to the purchaser's account.

7.4.1 To & Fro Conveyance Charges and Accommodation charges for the APTRANSCO Inspector attending Inspection and Testing should be borne by the bidder.

7.5 APTRANSCO Engineer will not inspect incomplete materials, if APTRANSCO Engineer returned without inspections of offered equipment due to non-readiness of material at their works, the financial loss to the APTRANSCO for deputing Engineer representative will be recovered from the concerned supplier.

7.6 In case of any discrepancy, APTRANSCO has the right to re test the Finished Goods (Type Test and Acceptance Test) at the Cost of Bidder.

8.0 QUALITY ASSURANCE PLAN:

8.1 The bidder shall invariably furnish the following information along with his offer failing which the offer shall be liable for rejection. Information shall be separately given for individual type of equipment offered.

i) The Structure of organization

ii) The duties and responsibilities assigned to staff ensuring quality of work

iii) The system of purchasing, taking delivery and verification of materials

iv) The system for ensuring quality of workmanship

v) The quality assurance arrangements shall confirm to the relevant requirement of ISO-9001 or ISO 9002 as applicable.

vi) **Statement giving list of important raw materials, names of sub-supplies for the raw materials, list of standards according to which the raw material are tested, list of tests normally carried out on raw material in the presence of suppliers representative, copies of test certificates.**

vii) Information and copies of test certificates as on (i) above in respect of bought out items.

viii) List of manufacturing facilities available

ix) Level of automation achieved and list of areas where manual processing exists.

x) List of areas in manufacturing process, where stage inspections are normally carried out for quality control and details of such tests and inspections.

xi) List of testing equipment available with the bidder for final testing of equipment specified and test plant limitation, if any vis-a-vis the type. Special acceptance and routine tests specified in the relevant standards. These limitations shall be very clearly brought out in "Schedule of Deviations" from the specified test requirements.

8.2 The Contractor shall within 30 days of placement of order, submit the following information to the purchaser.

i) List of the raw material as well as bought out accessories and the names of sub-suppliers selected from those furnished along with the offer.

ii) Type test certificates of the raw material and bought out accessories if required by the purchaser.

iii) Quality Assurance Plan (QAP) with hold points for purchasers inspection. QAP and purchasers hold points shall be discussed between the purchaser and contractor before the QAP is finalised.

The contractor shall submit the routine test certificates of bought out accessories and central excise asses for raw material at the time of routine testing if required by the purchaser and ensure that the quality assurance requirements of specification are followed by the sub-contractor.

8.3 **The Quality Assurance Programme shall give a description of the Quality System and Quality Plans with the following details.**

i. Quality System:

- The Structure of the Organisation
- The duties and responsibilities assigned to staff ensuring quality of work.
- The system of purchasing, taking delivery of verification of materials.
- The system of ensuring of quality workmanship.

- The system of control of documentation.
- The system of retention of records.
- The arrangement of contractor internal auditing.
- A list of Administrator and work procedures required to achieve contractors quality requirements. These procedures shall be made readily available to the purchaser for inspection on request.

ii) Quality Plans:

- An outline of the proposed work and program sequence.
- The structure of contractors organizations for the contract.
- The duties and responsibilities ensuring quality of work
- Hold and notification points.
- Submission of engineering documents required by this specification.
- The Inspection of the materials and components on request.
- Reference to contractors work procedures appropriate to each activity.
- Inspection during fabrication/ construction.
- Final inspection and test.

8.4 Training of Personnel:

The contractor shall provide necessary facilities for training personnel at their works/principals works relating to testing , operation and maintenance for four personnel free of cost. However, travel and incidental charges of the personnel will be borne by the purchaser.

9.0 DOCUMENTATION:

9.1 All drawings shall conform to International Standards Organization (ISO) 'A' series of drawing sheet/Indian Standards Specification. All drawings shall be in ink and suitable for micro filming. All dimensions and data shall be in S.I. Units.

9.2 List of Drawings and Documents:

The Bidder shall furnish four sets of the following along with his bid.

a) General outline and assembly drawings.

b) Graphics showing the performance of material.

c) General constructional features.

d) Type test reports in case the materials has been typed tested.

e) Test reports literature, pamphlets of the bought out items & raw materials.

9.3 The successful Bidder shall within 2 weeks of placement of order submit four sets of final version of all the above drawings for Purchaser's approval. The Purchaser shall

communicate his comments/approval on the drawings to the Supplier within two weeks of receipt of drawings. The supplier shall, if necessary, modify the drawings and resubmit four copies of the modified drawings for Purchaser's approval .The purchaser shall communicate approval with in two weeks of receipt of modified drawings. After receipt of Purchaser's approval, the Supplier shall, within three weeks, submit 20 prints and two good quality re-producibles of the approved drawings for Purchaser's use.

9.4 Six sets of the type test reports, duly approved by the purchaser, shall be submitted by the Bidder for distribution before commencement of supply. Adequate copies of acceptance and routine test certificates, duly approved by the purchaser, shall accompany the dispatch consignment.

9.5 The manufacturing of the equipment shall be strictly in accordance with the approved drawings and no deviation shall be permitted without the written approval of the Purchaser. All manufacturing and fabrication work in connection with the equipment prior to the approval of the drawing shall be at the Bidder's risk.

9.6 The Bidder shall also furnish twenty copies of bound manuals covering erection, commissioning, operation and maintenance instructions in English Language and all relevant information and drawings pertaining to the main equipment as well as auxiliary devices. Marked erection drawings shall identify the component parts of the equipment as shipped to enable Purchaser to carryout erection with his own personnel. Each manual shall also contain one set of all the approved drawings, type test reports as well as acceptance reports of the corresponding consignment dispatched.

9.7 Approval of drawings by the Purchaser shall not relieve the supplier of any of his responsibility and liability for ensuring correctness and correct interpretation of the drawings for meeting the requirements of the latest revision of the applicable standards, rules and codes of practices.

9.8 Raw Materials and bought out Items should be approved by APTRANSCO for utilization in manufacture of Battery chargers.

The equipment shall conform in all respects to high standards of engineering, design, workmanship and latest revisions of relevant standards at the time of supply and Purchaser shall have the power to reject any material which, in his judgment, is not in full accordance therewith.

10.0 PACKING, FORWARDING AND UNLOADING:

The equipment shall be packed in suitable crates so as to withstand handling during transit. The supplier shall be responsible for any damage to the equipment during transit, due to improper and inadequate packing and handling. The easily damageable materials shall be carefully packed and marked with the appropriate caution symbols. Wherever necessary, proper arrangement for lifting, such as lifting hooks etc., shall be provided. Any material found short inside the packing cases shall be supplied by the Supplier without any extra cost.

Each consignment shall be accompanied by a detailed packing list containing the following information:

a) Name of the consignee.

b) Details of consignment.

c) Destination.

d) Total weight of consignment.

e) Handling and unpacking instructions.

f) Bill of material indicating contents of each package and spare material.

The Supplier shall ensure that the packing list and bill of material are approved by the Purchaser before dispatch.

The Supplier shall arrange for unloading the material / equipment at his cost.

11.0 QUANTITY AND DELIVERY REQUIREMENTS:

i) This is set out in the proforma given in Price Schedule and the prices shall be quoted accordingly.

ii) The scope of supply shall include supply of 25% extra-quantity of bolts, nuts, washers, split pins, cotter pins and such other small loose items free of cost.

12.0 SUPERVISION SERVICES:

The supplier shall provide services of qualified personnel for the supervision of erection, testing at site and commissioning the Battery chargers supplied at free of cost.

13.0 GUARANTEED TECHNICAL PARTICULARS:

The Guaranteed Technical particulars for the equipment being supplied shall be provided with the Bid as specified in this technical specification at Annexure-I**.** The Bids received without guaranteed technical particulars shall be treated as Non-responsive.

****

A N N E X U R E – 1

GUARANTEED TECHNICAL PARTICULARS

FOR BATTERY CHARGING EQUIPMENT SUITABLE FOR VRLA AND TUBULAR BATTERY

S.No.	Description	220V 100AH	220V 200AH
1	Type of rectifier		
2)	No. of units		
3)	Manufacturer's type designation		
4)	A.C. supply		
	a) Voltage (Volts)		
	b) Current (Amps)		
	c) No. of phases		
	d) Frequency, cycles/Sec		
	e) Power factor		
5)	Percentage taps provided on transformer		
6)	Rated D.C. output for		
	a) Boost Charger (KW)		
	b) Float Charger (KW)		

7) Rated D.C. output voltage for

a) Boost Charger (Volts)
b) Float Charger (Volts)

8) Rated D.C. output current for

a) Boost Charger (Amps)
b) Float Charger (Amps)

9) Rate of D.C. voltage control for

a) Boost Charger (Volts)
b) Float Charger (Volts)

10) D.C. output voltage regulation from no load to full load volts

11) Maximum ripple content %

S.No.	Description	220V 100AH

12) Maximum permissible temperature rise over an ambient temperature of 50 Deg.C

13) Overall efficiency

14) Unit dimensions: Width x Height x Depth mm

15) Unit shipping weight Kgs

16) Load limiting feature (Please attach characteristics showing variation of voltage with increase in load)

17) Recommended settings for thermostat

a) Operation temperature Deg. C

b) Reset temperature Deg. C

18) List of major accessories provided

19) Recommended spares

TECHNICAL SPECIFICATION OF DCDB PANEL for 100AH battery charger

1.0 SCOPE:

1.1 This Specification provides for design, engineering, manufacture, assembly, shop testing, inspection and testing before dispatch, packing, forwarding and delivery at Site of DCDB Panels for 100AH Battery charger

1.2 It is not the intent to specify completely herein all details of the design and construction of equipments. However, the equipment shall conform in all respects to high standards of engineering, design and workmanship and shall be capable of performing continuous commercial operation acceptable to the Purchaser, who will interpret in a manner the meanings of drawings and specifications and shall have the power to reject any material which in his judgment is not in accordance therewith. The equipment offered shall be complete with all components necessary for its effective and trouble -free operation. Such components shall be deemed to be within the scope of Bidder's supply, irrespective of whether these are specifically brought out in this specification and/or the commercial order or not.

2.0 STANDARDS:

All equipment provided under this specification shall in general conform to the latest revision/amendment etc. of the following:

Sl.No.	Standard	Title
1.	IS 3231 IEC 255 Part I to III BS 142	Electrical relays for power system protection
2.	IS 1248 IS 2419 BS 89	Indicating instruments
3.	IS-13010,IS 722(Part Ito IX) IEC 521 BS 3T	Energy meters
4.	IS 4237	General requirement of switchgear and control gear for voltages not exceeding 1 **kV**
5.	IS 375	Marking arrangement for switch gear, busbars, main connection and auxiliary wiring
6.	IS 8686	Specification for static

	IEC-255Part V&VI	protective relays and tests.
	IEC 801-4	
7.	IEC 337	Control switches (LV switching
	IEC 337-1	devices for control and
	IS 6875	auxiliary circuits
8.	IS 2516	MCCB sand load break switches having a break capacity of 30 **kV**A rms at 415 volts
9.	IS 4047	Fuse switching units
10.	IS 2208	HRC fuse links
11.	IS 2705	Current Transformers
	IEC-185	
12.	IS 3156	Potential Transformers
	IEC-186	
13.	IS 8623	Factory built assemblies of switch gear and control gear for voltages upto and including 1000V AC and 1200V DC.
14.	IEC 439	Low voltage switch gear and control gear assemblies
15.	IS 4237	General requirement of switch- gear and control gear for voltages not exceeding 1000V
16.	IS 2147	Degree of protection provided by enclosures for low voltage switch gear and Control gear
17.	IS 5082	Aluminum alloy bars
18.	IS 1897	Copper strips for electrical purpose
19.	IS 3618	Code of practice for phosphate
	IS 6005	treatment of iron and steel
20.	IS 4064	Air break switches, air break dis-connectors, air break disconnectors and fuse combination units for voltages not exceeding 1000V AC or 1200V DC

21.	IS 9224 Part II	HRC catridge fuses
22.	IS 5578	Guide for marking of insulated conductors.
23.	IS 2032	Graphical symbols Ref- abbreviations

Equipment meeting with the requirements of other authoritative Standards, which ensure equal or better performance than the standards mentioned above, shall also be considered. When the equipment offered by the Bidder conforms to other standards, salient points of difference between standards adopted and the standards specified in this specification shall be clearly brought out in the relevant schedule. Four copies of such standards with authentic translation in English shall be furnished along with the offer.

3.0 CLIMATIC CONDITIONS:

3.1 The equipment to be supplied under this specification shall be suitable for satisfactory operation under the following climatic conditions.

i)	Location	Andhra Pradesh
ii)	Max. ambient air temp. (deg.C)	50
iii)	Min ambient air temp. (deg.C)	0
iv)	Average daily ambient air temperature(deg.C)	35
v)	Maximum Relative humidity (%)	80
vi)	Average annual rainfall (mm)	925
vii)	Isoceraunic Level (Days/Annum)	50
viii)	Max. wind pressure (Kg/sq.m.)	200
ix)	Seismic level (Horizontal Acceleration)	0.3g
x)	Max. altitude above mean sea level (Metres)	1000
xi)	Average No. of thunder storms days/ annum.	40
xii)	Average No. of rainy days/ annum.	90
xii)	Average No. of Months Tropical monsoon conditions/ annum.	3

4.0 PRINCIPAL PARAMETERS: The technical details of various equipments such as power transformers, circuit breakers and instrument transformers that will be associated with the control and relay panels in APTRANSCO are to be known by the bidder.

5.0 D.C. DISTRIBUTION BOARD :

5.1 The D.C. Distribution boards shall be provided with 2Nos. bus-bars made out of electrolytic copper of adequate capacity duly tinned supported by bus-bar supporters and are placed at sufficient distance instead of side by side to avoid accidental short circuit.

These busbars will be connected to 220V D.C. 2 wire system consisting of 220V, 100AH batteries or battery Charger rated for 220V D.C., 10A output.

The D.C. distribution boards required under this specification are to feed control circuits, and other auxiliary DC equipment of the Substations. DCDB shall be Free Standing, Floor Mounted Type Sheet Steel of CRCA, Non- Compartmental Type.

5.2 The D.C. distribution boards shall be of rugged construction, designed for working satisfactorily under the severest operating and climatic conditions and shall conform to the relevant ISS of latest issue.

5.3 The D.C. distribution boards shall be complete with all accessories and fittings necessary for the efficient and satisfactory operation whether they are specifically mentioned or not in the specification. The D.C. distribution boards shall be provided with the necessary reputed make switches, fuses, or DC Circuit breakers in the incoming and outgoing circuits and the equipment offered shall be complete in itself.

5.4 The D.C. distribution board shall comprise the following components for 220V, 10A, 100AH.

The D.C distribution board shall consist of provision for two 220 V DC input sources with 2 Nos Copper bus bar per each input source and DC Bus coupler arrangement. The outgoing feeders of DCDB shall be equally distributed between two sources of DC supply

220V, 100AH

a. Double pole MCB/MCCB
   for incoming DC supply — 2 Nos.(63A)
b. Double pole MCB/MCCB for bus coupler — 1 Nos
c. Double pole 20 A MCB
   Outgoing feeders — 12 Nos
d. Double pole 10 A MCB
   Outgoing feeders — 24 Nos

5.5 All the D.C. MCB/MCCBs shall be mounted on the front of the panel. The operating handle shall protrude out of the panel.

5.6 The D.C. distribution Board shall be of simplex type, sheet steel cubicle, indoor type, floor mounted and self supporting. It shall be totally enclosed, completely dust tight, weather and vermin proof. Ingress protection of DCDB cabinet shall be IP 42

5.7 The D.C. distribution board shall be made out of suitable angles and M.S sheet of 16 SWG for the front, 18 SWG for the back, top and sides, and 14 SWG for the bottom. The D.C. distribution board height shall be 1800mm,depth shall be 600 mm and suitable

width to accommodate all components i.e., adequate clearance/space should be provided between item to item to facilitate easy handling at the time of attending on repairs. The D.C. distribution Board shall be provided with hinged type rear doors.

5.8 The internal wiring of the D.C. distribution board shall be carried out with PVC insulated FRLSH, 1100V grade stranded copper conductor. The control wiring shall be carried out with 2.5 Sq.mm copper conductors. All wiring for external connections shall be brought to 1100V grade single piece molded terminal blocks with barriers. Suitable terminal blocks with stud type terminals shall be provided. The cables used shall have ISI marking on them. Extra 20% spare terminal blocks shall be provided.

5.9 The DC distribution board shall be provided with indicating LED lamps to indicate DC supply ON position such as 10mm LEDs for incoming supply and bus coupler and 5mm LEDs for outgoing feeders, for all circuits. Provision shall be made for writing the Name of the Circuit and necessary stickers shall be provided.

5.10 Digital multi-function meter with RS 485 port shall be provided for measuring DC voltage and current for both DCDB incomers.

5.11 The DC distribution boards shall be complete in all respects with all internal wiring incoming and outgoing terminals, labels, earthing connections and required cable glands.

5.12 All the final drawings and BOM of panels shall be approved by Design wing of APTRANSCO.

5.13 The DCDB shall be offered complete with all parts that are necessary or usual for their efficient operation. Such parts shall be deemed to be within the scope of this specification, whether specifically mentioned or not.

5.14 All cubicles shall undergo tank cleaning process according to latest IS in vogue followed by powdered coating process. The painting shall be uniform appearance and semi glossy finish (visually) without blistering. There shall not be any peeling at intersecting points or joints.

5.15 Miscellaneous Accessories:

Plug point- 230 Volts, single phase, 50 Hz AC plug point shall be provided in the interior of each cubicle with "ON-OFF" Switch for connection of hand lamps.

Interior lighting: Each panel shall be provided with a LED BULB rated for 230 volts, single phase, 50 HZ supply for the interior illumination of the panel during maintenance. The fittings shall be complete with switch fuse unit and switching of the lighting shall be controlled by the respective panel door switch. **Toggle Switch shall be provided in addition to the Door Limiting Switch.**

ON/OFF Switch shall be provided in addition to Thermostat in the Space Heater Circuit.

Exclusive Fuse Protection shall be provided in the PS Circuit (6A HRC Fuse & Link), Panel Lighting Circuit (2A HRC Fuse & Link) and in the Space Heater Circuit (2A HRC Fuse & Link) in addition to MCB in the Incoming AC Circuit.

Supporting steel: All necessary foundation anchor bolts and other parts of supporting and fastenings, of the panels shall be supplied by the contractor.

5.16 Earthing:

All panels shall be equipped with an earth bus securely screwed. Location of earth bus shall ensure no radiation interference for earth systems under various switching conditions of isolators and breakers. The material and the sizes of the bus bar shall be atleast 25 x 3 mm copper flat.

When several panels are mounted adjoining each other, the earth bus shall be made continuous and necessary connectors and clamps for this purpose shall be included in the scope of supply of contractor. Provision shall be made for extending the earth bus bars to future adjoining panels on either side.

Provision shall be made on each earth bus bars of the panels for connecting Purchaser's earthing grid. Necessary terminal clamps and connectors for this purpose shall be included in the scope of supply of contractor. The wire or screens should be clearly bonded and earthed at the gland plate.

All metallic cases of relays, instruments and other panel mounted equipment shall be connected to the earth bus preferably by independent copper wires of size not less than 2.5 sq.mm. The colour code of earthing wires shall be green. Earthing wire shall be connected on terminals with suitable clamp connectors and soldering shall not be permitted.

Looping of earth connections which would result in loss of earth connection to other devices where the loop is broken shall not be permitted. However, looping of earth connections between equipment to provide alternative paths to earth bus shall be provided.

6.0 TESTS

The bidder shall carryout routine tests on all associated equipment as per relevant Indian Standards/IEC Standards. The bidder shall submit along with the bid type test reports for protective relays, instruments, switches and other devices of the type and class being offered to assess the suitability of the equipment. Type tests must have been conducted not earlier than Ten years as on the date of bid opening. The bids received without type test reports will be treated as non-responsive.

ALL panels shall be subjected to the following tests:

a. Verification of degree of protection as per IS:2147 (Type Test)

b. High voltage test (2 kV for 1 minute) or as per relevant IS.

c. Insulation resistance test

d. Electrical control, interlock and sequential operation test

e. Verification of wiring and general arrangement as per approved schematic/G.A. diagrams.

7.0 Inspection:

7.1 i All routine tests shall be conducted at the time of inspection and inspection shall be made at the place of manufacturers unless otherwise specifically agreed upon by the manufacturer and purchaser at the time of purchase. The manufacturer shall offer the inspector representing the purchaser all reasonable facilities without charges to satisfy him that the panels/materials are manufactured/supplied as per the specification/final approved drawings.

ii. The supplier shall give 15 days (for local supplier) advance intimation to enable the purchaser to depute his representative for witnessing acceptance and routine tests.

iii. The panels shall be manufactured and offered for inspection as per the final approved General Arrangement Drawing and schematic drawing which will be binding both on supplier and purchaser.

iv. The panels shall not be dispatched from its point of manufacture unless the same are satisfactorily inspected and tested.

v. The acceptance of any material shall in no way release the supplier of his responsibility for meeting all the requirements of the specification and shall not prevent subsequent rejection of such material later found to be defective.

vi. The To and fro travelling expenses and accommodation facility for inspection officers shall be borne by the supplier of material / Bidder.

8.0 Quality Assurance Plan:

8.1 The bidder shall invariably furnish the following information along with his offer failing which the offer shall be liable for rejection. Information shall be separately given for individual type of equipment offered.

i. The Structure of organization

ii. The duties and responsibilities assigned to staff ensuring quality of work

iii. The system of purchasing, taking delivery and verification of materials

iv. The system for ensuring quality of workmanship

v. The quality assurance arrangements shall conform to the relevant requirement of ISO-9001 and ISO 9002 as applicable.

vi. Statement giving list of important raw materials, names of sub-supplies for the raw materials, list of standards according to which the raw material are tested, list of tests normally carried out on raw material in the presence of suppliers representative, copies of test certificates.

vii. Information and copies of test certificates as on (i) above in respect of bought out items.

viii. List of manufacturing facilities available

ix. Level of automation achieved and list of areas where manual processing exists.

x. List of areas in manufacturing process, where stage inspections are normally carried out for quality control and details of such tests and inspections.

xi. List of testing equipment available with the bidder for final testing of equipment specified and test plant limitation, if any vis-a-vis the type. Special acceptance and routine tests specified in the relevant standards. These limitations shall be very

clearly brought out in "Schedule of Deviations" from the specified test requirements.

8.2 The Contractor shall within 30 days of placement of order, submit the following information to the purchaser.

i. List of the raw material as well as bought out accessories and the names of sub-suppliers selected from those furnished along with the offer.

ii. Type test certificates of the raw material and bought out accessories if required by the purchaser.

iii. Quality Assurance Plan (QAP) with hold points for purchasers inspection. QAP and purchasers hold points shall be discussed between the purchaser and contractor before the QAP is finalised.

The contractor shall submit the routine test certificates of bought out accessories and central excise asses for raw material at the time of routine testing if required by the purchaser and ensure that the quality assurance requirements of specification are followed by the sub-contractor.

8.3 The Quality Assurance Programme shall give a description of the Quality System and Quality Plans with the following details.

i. Quality System:

i) The Structure of the Organization

ii) The duties and responsibilities assigned to staff ensuring quality of work.

iii) The system of purchasing, taking delivery of verification of materials.

iv) The system of ensuring of quality workmanship.

v) The system of control of documentation.

vi) The system of retention of records.

vii) The arrangement of contractor internal auditing.

viii) A list of Administrator and work procedures required to achieve contractors quality requirements. These procedures shall be made readily available to the purchaser for inspection on request.

ii. Quality Plans:

a. An outline of the proposed work and program sequence.

b. The structure of contractors organizations for the contract.

c. The duties and responsibilities ensuring quality of work

d. Hold and notification points.

e. Submission of engineering documents required by this specification.

f. The Inspection of the materials and components on request.

g. Reference to contractors work procedures appropriate to each activity.

h. Inspection during fabrication/construction.

i. Final inspection and test.

9.0 Documentation:

9.1 The bidder shall furnish the dimensioned drawings along with the bid indicating the details like components, make, quantity, ranges of meters, relays etc., in case the bidder quotes different makes of relays shall furnish copies of wiring scheme and literature of all relays.

10.0 Packing and Forwarding:

10.1 Name plate (giving details of equipment) in all equipments (i.e. relays, breaker switches, indicating lamps, push buttons etc.) mounted on the panel shall be provided. Name plates of suitable size on metal/PVC acrylic/thick plastic sheets shall only be provided. Stickers in no case will be accepted. Large and bold name plates shall be provided for circuit/feeder designation. Name plates for AC fuses, DC fuses, resistors and other equipment mounted inside the panel shall be provided clearly indicating the purpose for which they are meant for.

Name plates of suitable size shall be provided on the top of the panel clearly indicating manufacturers name, purchase order number and date, year of manufacturer, serial number of the panel and type of panels. The panels shall be packed in suitable sized strong and weather resistant wooden cases / crates.

10.2 Suitable cushioning, protective padding or spacer shall be provided to prevent damage to or deformation of the components during transit and handling.

10.3 All packing cases shall be marked legibly and correctly with the suppliers name and consignee's address so as to ensure their safe arrival at their destination and to avoid possibility of goods being lost or wrongly dispatched on account of faulty or illegible marking.

11.0 Coordination:

11.1 The bidder shall coordinate in all respects with the manufacturers of associated equipment and shall freely exchange all relevant information to ensure satisfactory and economic overall design. No extra charge shall be payable for these services. The bidder is responsible for the performance and service support for all the bought-out items he has used in the panel.

11.2 Erection of equipment called for in this specification will be undertaken by the purchaser. The bidder shall furnish full detailed instructions, complete with drawings and leaflets to enable satisfactory erection and commissioning of the plant by the departmental staff. Such instructions should reach the purchaser sufficiently in advance of the receipt of the equipment in order that all the preliminary arrangement for the erection of the equipment can be made by the department in advance so that there will be no delay in regard to erection as soon as the equipment are received.

It is obligatory on the part of the bidder to make available services of their testing and commissioning engineers (with necessary testing equipment) to assist the APTRANSCO staff in commissioning their equipment as and when their services are requisitioned.

12.0 Tropical Treatment:

12.1 Under the climatic conditions prevalent at site, the equipment furnished under the specification will be subjected to operation under ambient temperatures of 50 deg. C and very high relative humidity. All equipment shall therefore, be suitably designed and treated for normal life and satisfactory operation under the hot and humid climatic conditions prevalent at site and shall be dust and vermin proof. All parts and surfaces which are subject to corrosion, shall be made of such material and shall be provided with such protective finishes as would protect the equipment installed from any injurious effects of excessive humidity. All electrical auxiliary equipment shall be specifically treated for tropical conditions, and the materials and methods for this treatment shall be approved in advance by the Purchaser.

13.0 Foundations and Fixtures:

Foundations will be prepared by the Purchaser in accordance with the basic designed data to be supplied by the contractor. All fixtures necessary for proper erection of the embedded parts shall be supplied by the supplier if required.

DCDB PANELS:

TECHNICAL PARTICULARS

A. Instruments (Digital Voltmeter and Ammeter)

a. Make
b. Type
c. Applicable standards
d. Range
e. Shape
f. Size
g. Scale Length (mm)
h. Finish
i. Temperature at which calibrated

B. Bus bars and Insulators

1. Material of bus bars and applicable standards
2. Cross sectional area of bus bars in Sq. mm
   i) For horizontal bus bar
   ii) For vertical bus bar

3. Type of bus bar insulation
   (Bare/Painted/Epoxy insulated
   sleeved)
4. Minimum clearance (in mm)
   a) Phase to phase
   b) phase to earth
5. Continuous current rating
   At 50 Deg. Ambient temperature
6. Short time current rating
   for 1 sec.
7. Material of the support
   Insulator

C. MCCBs/MCBs

1. Make and type
2. Rated current at design
3. Continuous current at
   site ambient temperature
4. Rated symmetrical
   breaking current
5. Rated making current
6. Short time current rating
7. Earth Bus
   1) Material
   2) Cross sectional area (Sq. mm)

D. Panels

1. Material and thickness of
   a) Front sheet (mm)
   b) Back sheet (mm)
   c) Barrier, sheet between
      units (mm)
   d) End covers (mm)
2. Dimension of Panels (L x H x D)

3. Symmetrical short circuit
   withstand current of
   switchgear cubicle in KA
4. Degree of protection as Per IS/IEC 60947
5. Width of cable alley
6. Total weight of the panel
   including all the enclosures
8. Painting (powder Coated)
   a) Colour
   b) Finish
   c) No. of coats

TECHNICAL SPECIFICATION

For 220V, 200AH Battery charging Equipment and D.C. Distribution boards

1 SCOPE:

This specification covers design, engineering, manufacture, stage testing, inspection and testing before supply and delivery at site/CIP destination of following Battery charging equipment and DC distribution boards suitable for the Maintenance Free Valve Regulated Lead Acid Battery sets and Tubular type Battery sets **(VRLA/TUBULAR)**.

220V 200AH Battery charging equipment comprising separate Float Cum Boost Charger & separate Float Charger suitable for 220 Volts 200AH Maintenance Free Valve Regulated Lead Acid type batteries and Tubular type Battery sets **with separate DC Distribution board** with dropper diode in separate enclosure.

2 STANDARDS:

The material shall conform in all respects to the relevant Indian Standard Specifications with latest amendments thereto.

Indian Standard	Title
IS 16539/2017 (Part1)	Semiconductor Converters.
IS 6619	Safety Code for Semiconductor Rectifier Equipment
IS/IEC 60947	Low-voltage switch gear and control gear.
IS/IEC 61558 similar	Safety of power transformers, power supply units and
IS/IEC 60269	Low-Voltage Fuses for Voltages not Exceeding 1000 V AC or 1500 V DC

Equipment conforming to other internationally accepted standards, which ensure equal or higher quality than the standards mentioned above, would also be acceptable. In case the Bidders who wish to offer material conforming to the other standards, salient points of difference between the standards adopted and the specific standards shall be clearly brought out in relevant schedule. Four copies of such standards with authentic English Translations shall be furnished along with the offer. In case of conflict the order of precedence shall be (i) IS, (ii) IEC, (iii) Other standards. In case of any difference between provisions of these standards and provisions of this specification, the provisions contained in this specification shall prevail.

3. CLIMATIC CONDITIONS:

The material offered shall be suitable for operation under the following Service conditions.

i)	Location	:	At various locations in the State of Andhra Pradesh
ii)	Maximum ambient air temperature (deg C)	:	50
iii)	Minimum ambient air temperature (deg C)	:	5
iv)	Average daily ambient Air temperature (deg.C)	:	35
v)	Maximum relative Humidity(%)	:	80
vi)	Maximum altitude above mean sea level (Metres)	:	1000
vii)	Average annual rain fall(mm)	:	940
viii)	Maximum wind pressure (kg/Sq.mm)	:	200
ix)	Isoceraumic level(days per year)	:	50
x)	Seismic level(Horizontal accn.)	:	0.10 g.
xi)	Noise level	:	45 db

The Battery Chargers and D.C. Distribution Boards shall be for use in moderately hot and humid tropical climate, conducive to rust and fungus growth.

4. PRINCIPAL PARAMETERS:

The equipment shall conform to the following specific parameters:

Sl.No.	Item	Specification
1.	Type of Installation	In door

2.	AC voltage	415 V +20% to –20% (for 220 V batteries)
3.	No. of phases	Three
4.	Frequency	50 c/s ± 5%
5.	Regulation	± 1%
6.	Ripple content	2% RMS without Battery connected

5.0 **TECHNICAL REQUIREMNTS:**

5.1 The Battery charging equipment shall comprise of a Float Charger and a Float Cum Boost Charger.

General remarks:

1. The following are the parameters to be displayed in Multi-Function digital meters
AC Voltage parameters: AC input voltage and currents (All three phases), power and frequency-1 No. may be displayed in digital multi-function meter. **All AC power supply wires should have colour coding i.e. Red, Yellow, Blue, and Black.**

   DC Voltage parameters: DC Voltage and currents (FC, FCBC, Load, Battery)-4 Nos. 2 Line display meters individually for FC, FCBC, Load and Battery. **All DC power supply wires should have colour coding i.e. Red for Positive and Black for Negative.**

   The above parameters are to be communicated to SCADA through Modbus over serial port (RS485)/Ethernet port with RS 485 to Ethernet converter.

2. The following are the parameters to be displayed in Analog meters Earth Leakage current, Battery current (Indicating whether charging/discharging).
The Earth leakage current is to be communicated with SCADA by providing a Transducer with 4-20 milli amp output.

3. All the alarm, trip and status signals are to be communicated to SCADA through Modbus over serial port (RS 485)/Ethernet port. An annunciator panel shall be provided for this purpose.

4. For setting of Float/boost voltages, separate potentiometers are to be provided for both VRLA and Tubular type batteries i.e. one for setting float voltage in FC mode for VRLA, one for setting float voltage in FC mode for Tubular, one for setting float voltage in FCBC mode for VRLA, one for setting float voltage in FCBC mode for Tubular, one for setting boost voltage in FCBC mode for VRLA, one for setting boost voltage in FCBC mode for Tubular.

5. Short circuit protection shall be provided. **The charger shall withstand during short circuit condition. The same shall be demonstrated at the site during commissioning along with other logics and protections and same shall be recorded in MOM.**

The charger shall either limit the current or turn off the gate pulses of the charger during short circuit condition.

6. Kick fuses /Spring-loaded fuses are not allowed.
7. Terminal blocks shall be of ELMEX-KABTM4 or M/s Connect well STH4 .
8. All internal wiring to be routed through cable trunk.
9. Provision for accept/reset of annunciation from Remote maybe arranged. (or) **Potential Free Contacts shall be provided for Alarm Accept & Alarm Reset.**
10. Provide rectifier fuse with base and carrier.
11. Provide internal General arrangement diagram.
12. The Positive and negative DC bus bars maybe separated from each other by approx. one foot.
13. For Earth fault protection, a separate relay shall be used which can be replaced easily without disturbing any other circuits.
14. DCDB should be supplied separately.
15. **3 Pole Battery isolator switch/MCCB is to be provided in Battery circuit which shall be suitable for withstanding 30 minutes battery discharge current rate and also MCB/MCCB of battery output rating not less than 63 Amps for 100AH, 100 Amps for 200AH and 200 A for 400 AH battery sets.**

    The power circuit, MCBs, cables etc between battery terminals to the load including the cable connecting charger to DCDB and also the internal ckts of DCDB shall be adequately designed to meet 30 minutes of battery discharge current and through short circuit current produced by the battery bank, and other circuits shall be as per the charger rating.
16. **All internal components are to be mounted on channels for easy replacement. All internal power circuits (DC/AC) shall be adequately designed to meet 30 minutes of Battery discharge current and through short circuit current produced by the battery bank.**
17. **The selection of modes like AUTO/MANUAL, VRLA/Tubular, FC/FCBC shall be done through switch only. Auto / Manual selection and VRLA / NVRLA selection switches (3 No.) shall have a lock & key facility to avoid mal-operation.**
18. **Battery current limiter shall be provided and set as per C10 rating of battery set and also having the provision for adjusting the battery current limiter setting.**
19. **There shall not be any Auto Mode in Charger for NVRLA/Tubular Battery Units. If the operator unknowingly select this mode, the Charger shall operate in Manual Mode only. After selecting NVRLA battery mode the control logics of VRLA mode shall be isolated from charger circuit.**

    The following are the details of Float / Boost voltages for Batteries.

	VRLA type	Tubular type

Float voltage set point(per cell)	2.05V to 2.3 V.	2.05V to 2.3 V.
Boost Voltage set point(per cell)	2.15V to 2.4 V.	2.15V to 2.7 V.
Float voltage range	190 V to 250 V	190 V to 250 V
Boost voltage range	240 V to 265 V	240 V to 305 **V**

5.1.1 FLOAT CHARGER:-

The Float Charger is to operate on three phase A.C. supply of 415 Volts with +20% to –20% variation, 50 C/S. The Float Charger shall be capable of Floating the Battery i.e., trickle charge the Battery comprising of 110Nos., 2Volts, 200AH lead acid cells at about 2.23v per cell and simultaneously supplying a constant load of **25 Amperes** in an average Ambient temperature of 35 Deg. C. The Charger shall be provided with a regulator to facilitate controlling of the cell voltage and to stabilize the output voltage within + or - 1% of the set D.C. value, for A.C mains supply voltage variation of +20% to –20% and frequency variation of + or - 5% and D.C. load variation of 0-100% and also when all the three variations occur Simultaneously.

The Float Charger shall be provided with an automatic current limiting facility such that when Float Charger output current exceeds 10% above the rated current, the Float Charger voltage should be brought down automatically so that the Float Charger output current does not exceed the set value. The Float Charger shall be of 3 phase full wave full-controlled thyristor bridge rectifier type with automatic voltage regulator unit and with necessary printed circuit boards, transformers and relays etc.

5.1.2 FLOAT CUM BOOST CHARGER: -

The Float Cum Boost Charger is to operate on a three phase supply of 415 Volts with + 20% -20% variation 50 C/S. The Float Cum Boost Charger shall be capable of Boost charging the battery comprising of 110 Nos. 2V 200AH lead acid cells upto a maximum cell voltage of 2.7v per cell at a maximum charging current of **50 Amperes** for the Chargers suitable for 200AH batteries in maximum ambient temperature of 50^0 C. The Float Cum Boost Charger shall be capable of Boost charging the battery comprising of 110 Nos., 2Volts, 200AH lead acid cells from fully discharged condition to fully charged condition as per C10.

5.2 **The Battery charging equipment shall be complete with all parts that are necessary for their efficient operation. Such parts shall be deemed to be within the scope of this specification whether specifically mentioned or not. Diode dropper circuit shall be provided in separate enclosure.**

5.3 OPERATION

- A Charger should have 2 No independent charging circuits such as one for FC and one for FCBC
- The charger should have AC over voltage protection relay with alarm and trip with time setting facility.
- Each charging circuit should be **3-ph full wave full controlled** bridge rectifiers (6 thyristors) for smooth DC wave form

- Both charging circuits should be of constant voltage type with soft start and should have short circuit protection with current limiters.
- Each charging circuit should maintain constant DC set voltage (± 1%) with AC voltage regulation (+20% to -20%)
- Each charging circuit should maintain constant DC set voltage (± 1%) up to rated current limit set and above which the voltage should fall accordingly (drooping characteristic).
- 1st charging circuit named as **Float Charger (FC)** should always be ON and support the DC load connected to Charger along with trickle charging the battery.
- 2nd charging circuit named as **Float Cum Boost Charger(FCBC)** should be generally in OFF condition, up on raise of battery current drawn above a specified value the FCBC circuit should ON automatically by detaching battery from Float charging circuit through a DC contactor and should charge the batteries in high rate mode i,e BOOST mode. And up on full charging of batteries with the decrease of battery current drawl below the specified value FCBC should switch OFF with battery again connecting to Float Charger through DC contactor.
- In case of failure of FC, the FCBC should ON and function as Float Charger supplying DC loads with trickle charging the battery.
- FCBC should work as Boost Charger in case of main FC healthy and should act as Float Charger in case of FC failure thus fulfilling the name **Float Cum Boost Charger (FCBC)**.
- Charger should have **Auto/Manual Float/Manual Boost** mode of operation in VRLA type and charger should have **Manual Float/Manual Boost** mode of operation only for Tubular Type and separate mode selector switches shall be provided for VRLA type and Tubular type each and Battery type selector switch shall be provided for selecting VRLA Type or Tubular Type.
- In Auto mode (For VRLA only)
    - FC should switch ON automatically with AC supply ON
    - FCBC should ON automatically, for BOOST charging the battery, with increase of battery current and should OFF with decrease of battery current provided FC healthy only.
    - FCBC should ON automatically in case of failure of FC and should work as Float Charger and should OFF upon FC becomes healthy.
- In Manual Float mode (for VRLA type)
    - FC should remote OFF and FCBC should ON and function as Float charger for float charging the Battery and DC Contactor ON for Manual float selection of VRLA mode.
    - If FCBC OFF/Trip, FC should ON and function as FC charger.
- In Manual Boost mode(for VRLA)
    - If FC is healthy and supporting DC loads, the FCBC should ON and function as Boost Charger, for BOOST charging the battery and DC Contactor OFF.

- If FC is FAIL/OFF/Trip, the FCBC should only work as Float charger and DC contactor ON.

- In Manual Float mode(for Tubular)
  - If FC is healthy and ON for supporting DC loads, the FCBC should ON and function as float Charger, for float charging the Battery and DC Contactor OFF.
  - If FC is FAIL/OFF/Trip, the FCBC should only work as Float charger and DC Contactor ON.
- In Manual Boost mode(for Tubular)
  - If FC is healthy and ON for supporting DC loads, the FCBC should ON and function as Boost Charger, for Boost charging the Battery and DC Contactor OFF.
  - If FC is FAIL/OFF/Trip, the FCBC should only work as Float charger and DC Contactor ON.
- The DC contactor connecting battery to Float Charger/Loads should OFF only while **FC is healthy** & **FCBC is ON** either in Auto mode or Manual float/Manual Boost mode. The DC contactor shall follow as per the logic table given below.
- While FC feeding loads and FCBC is Boost charging the battery exclusively with DC contactor open, upon immediate failure of Ac supply/FC fail, the DC contactor should close immediately for delivering full DC voltage.
- The following logic matrix should be followed for the operation of charger.
- **The Manufacturer should submit the Logic (shall explain how they are providing all the protections as per APTRANSCO Specification) of all Protections available for the Battery Charger at the time of Applying for Drawing Approvals. The Approved Logic Drawing shall be Stickered, Painted on inside of the Front Door.**

LOGIC MATRIX FOR FC AND FCBC OPERATIONS (For VRLA mode)

System condition	Mode selection	Condition of FC	Condition of FCBC	Battery current (Set value)	Interlocking Contractor (K1)	Load	Battery
AC Supply Available	AUTO	Healthy	Healthy	Less than set value	ON	Supplied by FC (Float voltage)	On Float Voltage (By FC) and FCBCshall be Automatically OFF
AC Supply Available	AUTO	Healthy	Healthy	Greater than set value	OFF	Supplied by FC (Float voltage)	On Boost Voltage (By FCBC)
AC Supply Available	AUTO	FAIL	Healthy	Less/ Greater than set value	ON	Supplied by FCBC (Float voltage)	On Float Voltage (By FCBC)
AC Supply Available	AUTO	Healthy	FAIL	Less/ Greater than set value	ON	Supplied by FC (Float voltage)	On Float Voltage (By FC)
AC Supply Available	MANUAL FLOAT	Healthy	Healthy	Less/ Greater than set value	ON	Supplied by FCBC (Float voltage) and FC shall be Automatically OFF	On Float Voltage (By FCBC)
AC Supply Available	MANUAL BOOST	Healthy	Healthy	Less/ Greater than set value	OFF	Supplied by FC	On Boost Voltage (By FCBC)
AC Supply Available	MANUAL FLOAT/ BOOST	FAIL	Healthy	Less/ Greater than set value	ON	Supplied by FCBC (Float voltage)	On Float Voltage (By FCBC)

AC Supply Available	MANUAL FLOAT/ BOOST	Healthy	FAIL	Less/ Greater than set value	ON	Supplied by FC (Float voltage)	On Float Voltage (By FC)
AC Supply Available	ANY MODE	FAIL	FAIL	Less/ Greater than set value	ON	Supplied by Battery	On Discharge
AC Supply Fail	ANY MODE	Healthy	Healthy	Less/ Greater than set value	ON	Supplied by Battery	On Discharge

LOGIC MATRIX FOR FC AND FCBC OPERATIONS (For TUBULAR mode)

System condition	Mode selection	Condition of FC	Condition of FCBC	Interlocking Contractor (K1)	Load	Battery
AC Mains Available	MANUAL FLOAT	Healthy	Healthy	OFF	Supplied by FC(Float voltage)	On Float voltage (By FCBC)
AC Mains Available	MANUAL FLOAT	Healthy	FAIL	ON	Supplied by FC (Float voltage)	On Float Voltage (By FC)
AC Mains Available	MANUAL FLOAT	FAIL	Healthy	ON	Supplied by FCBC(Float voltage)	On Float voltage (By FCBC)
AC Mains Available	MANUAL BOOST	Healthy	Healthy	OFF	Supplied by FC (Float voltage)	On Boost Voltage (By FCBC)
AC Mains Available	MANUAL BOOST	Healthy	FAIL	ON	Supplied by FC (Float voltage)	On Float Voltage (By FC)
AC Mains Available	MANUAL BOOST	FAIL	Healthy	ON	Supplied by FCBC (Float voltage)	On Float Voltage (By FCBC)
AC Mains Available	ANY MODE	FAIL	FAIL	ON	Supplied by Battery	On Discharge
AC supply Fail	ANY MODE	Healthy	Healthy	ON	Supplied by Battery	On Discharge

5.4 **The Float Charger for 220 Volts,200AH batteries shall comprise of the following components but not limited to them.**

i)	AC 3-pole Moulded Case Circuit Breaker of suitable capacity	:	1 No.
ii)	Pilot (Clustered LEDs) lamps to indicate A.C. mains supply	:	3 Nos. (Common for both FC&FCBC)
iii)	3 phase full wave Full-controlled thyristor bridge rectifier with fuses	:	1 No.
iv)	Automatic voltage regulator unit with necessary printed circuit boards transformers and relays, etc.	:	1 Set
v)	Filter choke and filter condenser	:	1 Set
vi)	Double wound impregnated natural air cooled three phase mains transformer of suitable rating	:	1 No.
vii)	D.C. 2 pole Moulded Case Circuit Breaker of suitable capacity	:	1 No.
viii)	A.C Power contactor and thermal overload relay	:	Each one number

Though not specifically included in the above list, the supplier of the battery Chargers shall supply all the required control equipment for trouble free operation of the Chargers..

5.5 The Boost Charger for 220 volts, 200AH batteries shall comprise of the following components but not limited to them.

i)	AC 3-pole Moulded Case Circuit Breaker of suitable capacity	: 1 No.
ii)	Pilot (Clustered LEDs) lamps to indicate A.C. mains supply	: 3 Nos. (Common for both FC&FCBC)
iii)	3 phase full wave Full-controlled thyristor bridge rectifier with fuses	: 1 No.
iv)	Automatic voltage regulator unit with necessary printed circuit boards transformers and relays, etc.	1 Set
v)	Filter choke and filter condenser	: 1 Set
vi)	Double wound impregnated natural air cooled three phase mains transformer of suitable rating :	1 No.
vii)	D.C. 2 pole Molded Case Circuit Breaker of suitable capacity	: 1 No.
viii)	A.C Power contactor and thermal overload relay	- Each one number

5.6 The battery Chargers shall also comprise the following equipment.

i) 1 No. silicon Tap Cell diode connected to 107th cell for VRLA Battery and 93rd Cell for Tubular Battery and the D.C. positive bus to maintain continuity of D.C. supply to the D.C bus in the event of A.C. failure while Boost Charger in service and to avoid interruption short time of to connect 1st cell and the 107th cell for VRLA Battery and 93rd cell for Tubular Battery positive.

ii) Ripple filtering devices.

iii) 1 No. triple pole Molded Case Circuit Breaker of suitable capacity for connecting to 110th cell positive and 1st cell negative and 107th cell for VRLA Battery and 93rd cell for Tubular

iv) Internal light operated on 230 Volts single phase, 50 C/S, A.C. system with door opening/closing ON/OFF switch.

v) 2 Nos. earthing lugs suitable for receiving 120 sq.mm copper conductors.

vi) Space heaters suitable for operation AC 230 V, 1 Amp, 50 c/s.

vii) Heater ON/OFF switch.

viii) Undrilled Gland plate for all the external cables.

5.7 The Chargers shall have built-in automatic voltage control and load limiting features. The voltage regulator shall automatically sense, monitor and regulate the D.C. voltage to within + 1% of the set value from No load to full load and under A.C. input supply voltage and frequency fluctuations. Load limiting features shall automatically reduce the output voltage of the Charger on loads more than the rated load.

5.8 The ripple content of the Charger D.C. output shall not exceed 2% when the battery is not connected.

5.9 Alarm and fault indicating lamps shall be arranged in the following faults. **(**Facia Display Unit) with RS-485 port will be provided

a) Main incoming AC supply "ON"

b) DC supply "ON"

c) A.C. input Circuit Breaker fail both Float and Float Cum Boost Charger.

d) D.C. output Circuit Breaker fail both Float and Float Cum Boost Charger

e) Mains failure

f) Phase failure

g) D.C. output over voltage

h) D.C Under voltage

i) Rectifier fuse failure for both Float Charger and Float Cum Boost Charger

j) Filter condenser fuse failure for both Float Charger and Float Cum Boost Charger

k) Battery earth fault.

l) Battery MCCB Fail

m) DC contactor OFF

n) AC over voltage Alarm

o) AC under voltage Alarm

Battery current Limiter: The charger should have battery current limiter to limit the Battery charging current upto C10 Charging current with Battery current limit setting provision.

Load Limiters:

All chargers shall have load limiters which shall cause, when the voltage control is in automatic mode, a gradual lowering of the output voltage when the DC load current exceeds the load limiter setting of the Charger. The Load-limiter characteristics shall be such that any sustained overload or short circuit in DC System shall not damage the Charger.

Soft Start:

The charger shall be provided with soft start feature so as to make DC output voltage to rise gradually to its rated value over duration of 6-8 seconds whenever the charger is powered or at the time of restoration from shutdown. Soft start feature should be invariably provided to minimize the inrush current from the float charger to discharged battery.

Battery over voltage protection

Over voltage shutdown/cutback: The charger should compare DC output voltage with over voltage set point, above which the charger should inhibit the firing pulse to charger

thyristors so as to shut down the system thus avoiding over voltage at charger output. The same can be reset through over voltage reset in set parameters.

Filter Circuit

Suitable filter circuits shall be provided in the chargers. The ripple content (peak to peak) in the output voltage shall not exceed 2%, irrespective of the DC load.

Contactors

All battery chargers shall be provided with AC contactors on the input side, shall be of air break type and suitable for continuous duty. The operating coil shall be rated for 415 Volts +/- 20% AC

AC Contactor shall be provided after the secondary of the rectifier transformer. Thermal overload relay shall control the contactor.

Thermal Overload Relay

A thermal overload relay incorporating a distinct single phasing protection (using differential movement of bimetal strips) shall also be provided for the AC input.The relay shall trip the above contactor.

Rectifier Transformer

The rectifier transformer shall be copper wound Vacuum Impregnated continuously rated and natural air cooled (N.A.) type. The rating of the rectifier, transformer shall correspond to the rating of the associated rectifier assembly. The Bidder shall provide the supporting calculations for the transformer capacity during design & detailed engineering. Temperature rise of the transformer during fully loaded condition shall not cause damage to the electronic components of the charger. Necessary ventilation arrangements have to be made to limit the temperature rise to safe limits. There shall be separate transformers for FC charger and FCBC charger, physically located at the bottom of the panel for easy inspection.

Rectifier Assembly

Phase rectifier bridge assembly shall be full wave full controlled (6 thyristors) shall be provided for both float and float cum boost charger. The rectifier assembly shall be designed to meet the duty as required by the respective Charger. The rectifier shall be provided with their own heat sink having their own dissipation arrangements with natural air-cooling. Necessary surge protection devices and semi-conductor fuses shall be provided in each arm of the rectifier connections. It should be possible to easily replace individual thyristors. Each thyristor shall be fixed with bolted arrangement for easy testing and replacement.

POTENTIAL FREE CONTACTS

The following Potential Free Contacts shall be available on FC & FCBC:

i. AC Mains Fail
ii. FC AC I/P Circuit Breaker Trip
iii. FCBC AC I/P Circuit Breaker Trip
iv. FC DC O/P Circuit Breaker Trip

v. FCBC DC O/P Circuit Breaker Trip
vi. Battery Earth Fault
vii. FC Rectifier Fuse Fail
viii. FCBC Rectifier Fuse Fail
ix. FC Filter Fuse Fail
x. FCBC Filter Fuse Fail
xi. DC Over Voltage
xii. DC Under Voltage
xiii. Battery Input Circuit Breaker Trip/Off
xiv. Group alarm
xv. Alarm Accept
xvi. Alarm Rest
xvii. DC Contactor OFF
xviii. Phase Fail
xix. AC over voltage Trip
xx. AC under voltage Trip

5.10 Necessary PCBs/relays for the above shall be provided wherever necessary.

5.11 A block diagram to indicate the general protection features provided for the Charger shall be furnished along with the Bid.

5.12 Necessary relays for annunciation for faults mentioned in 5.9 shall be provided. They shall be suitable for operation of 220 V D.C. supply with + 15% to –15% variation. Necessary circuitry for canceling alarm shall be provided. On canceling the alarm, the sound only should go and lamp should continue till the fault is rectified/cleared.

5.13 A Blocking diode of adequate rating shall be provided in the positive pole of the output circuit of each charger to prevent current flow from the DC Battery into the charger.

5.14 In the event of A.C supply failure, when Battery is on Boost charge, the available battery capacity shall be automatically connected to the D.C Bus

5.15 All fuses shall be HRC non-deteriorating type wherever necessary. Unless otherwise specified, the fuses shall be of class-4 (80KA prospective breaking current) for A.C. circuits and class-2(33 KA prospective D.C. current) for D.C. circuits.

MCB/MCCBs

i. AC input and DC output MCB/MCCBs shall be provided to in all chargers in addition to contactor and thermal over load relay and they shall be of heavy duty single through group operated quick make, quick break type complying with IS: 8828 & IEC: 898-1987 and suitable for 400/240 V A.C and 220 V DC. All MCCBs shall be suitable for continuous duty. MCCBs should have auxiliary contacts for annunciation.

ii. MCB/MCCBs shall be so designed with suitable breaking capacity in conformity with above IEC & ISI. The bidder shall ensure that all MCCB/MCBs are adequately rated so as to be fully protected during all abnormal operating conditions.

iii. MCCB/MCB shall be flush mounted on Battery Chargers Panels.

iv. Suitable door inter locking lamps shall be provided with micro switches.

v. MCCB shall be provided with bimetallic & magnetic type release for over current & short circuit protection. The setting of the thermal release shall be adjustable between 50% to 100% of the rated current.

vi. MCBs for A.C and D.C control supply and heaters wherever required shall be mounted inside the chargers.

5.16 The indicating instruments shall be of flush mounting type with dust tight covers. The instruments shall be approximately 96 mm square with 240 deg. C scale or alternatively 144 mm square with 90deg.C scale. Dials shall be white with black numbers and lettering.

5.17 **Transducers:** The contractor shall supply the Transducers for the measurement of Earth Leakage Current. The Output (4-20mA) of the transducers shall be carried up to Terminal Blocks for taking them to Auxiliary BCU for Automation.

5.18 The internal wiring of the Charger shall be carried out with PVC insulated 1100 V FRLSH grade stranded copper conductor. The control wiring shall be carried out with 2.5 sq.mm copper conductors. All wiring for external connections shall be brought to 1100 V grade single piece mounted terminal blocks with barriers. Suitable vertical terminal blocks shall be provided. Terminal connectors provided for wiring shall be stud type and not screw type.

5.19.1 All the external cables to be connected to the Charger shall be arranged for bottom entry.

The Float charging equipment and Boost charging equipment shall be mounted in a metal enclosed, sheet steel cubicle, indoor floor mounting, free standing type. It shall be totally enclosed, completely dust tight, weather and vermin proof. The cubicle shall be adequately ventilated with louvers to facilitate the cooling of transformers and rectifiers. The ventilating openings shall be less than 3 mm size. The panel shall be complete with internal wiring, terminal board for internal and external connections. The panel shall be made out of suitable angles and M.S. Sheet of 16 SWG for the front,18 SWG for the back, top and sides, and 14 SWG for the bottom. The height of the panel shall be 1800 mm and depth 600 mm and panel shall have suitable width to accommodate the charging equipment. i.e., adequate clearance / space should be provided between item to item in the Charger, to facilitate easy handling at the time of

attending on repairs. The Charger shall be provided with hinged type rear doors. Ingress protection of Charger shall meet with IP 42.

Each Charger shall be furnished completely wired upto power cable lugs and terminal blocks with insulated barrier between each terminals of AC Input, DC output and Battery input and ready for external connections. The control wiring shall be carried out with 1.1 KV PVC insulated, 2.5 sq.mm. Stranded copper wires, Control terminals shall be suitable for connecting two wires, with 2.5 sq.mm stranded copper conductors. All terminals shall be numbered for ease of connections and identification. Each wire shall bear a ferrule or tag on each end for identification. At least 20% spare terminals shall be provided for control circuits.

The insulation of all circuits except the low voltage electronic circuits shall withstand test voltage of 2 kV AC for one minute. An air clearance of at least ten (10) mm shall be maintained throughout for such circuits, right up to the terminal lugs. Whenever this clearance is not available, the live parts shall be insulated or shrouded.

The low voltage electronic circuits tests shall be as per relevant standards and the bidder has to furnish supporting documents for low voltage electronic circuits.

The battery Chargers shall be tested for its ratings before dispatch.

5.20 ACCESSORIES AND SPARES REQUIRED

The following accessories shall be supplied along with each battery Charger.

i)	Suitable wrenches	:	2 Nos. per Set.
ii)	HRC Fuses, Glass Fuses, MCCBs, Thyristors and such small parts shall be provided 25% extra as spare.		
iii)	Battery Charger O&M Manual along with drawings	:	2 Nos.

5.21 The Battery Charger shall be offered complete with all parts that are necessary or usual for their efficient operation. Such parts shall be deemed to be within the scope of this specification, whether specifically mentioned or not.

5.22 **PAINTING**: Regarding painting of the panel, Bidder may please note that outside visible portion of the panel (when doors are closed) must be painted with light gray shade No.631 of IS-5 and inside of the panel with Brilliant white and Base channel with black paint after necessary primary coats of red-oxide.

6.0 TESTS :

6.1 **Type Tests :**

6.1.1 All the offered equipment shall be fully type tested by the bidder as per the

relevant standards including the type tests mentioned below. The type tests must have been conducted on the similar or higher capacity from recognized test laboratory within 7 years prior to the date of bid opening. The bidder shall furnish two sets of type tests reports as per the relevant standards for each equipment offered along with bid. The offers received without these type tests reports shall be treated as non -responsive.

a) Voltage Regulation Test
b) Load limiter characteristics Test
c) Efficiency Tests
d) High Voltage Tests
e) Temperature rise Test
f) Short circuit Test at no load and full load at rated voltage for sustained short circuit
g) Degree of protection Test
h) Measurement of ripple by oscilloscope

Additionally, If the type tests submitted are of different rating from the offered rating, then ,all the above tests except the Ingress Protection test shall be carried in-house at factory premises on one charger in the presence of APTransco

6.1.2 Routine Tests: Each equipment manufactured shall be tested for (i) Insulation Resistance Test (ii) High Voltage Test (iii) Annunciation Verification (iv) DC output voltage regulation (v) Efficiency at full load (vi) Power factor (vii) DC output voltage ripple (viii) Battery path current limit and (ix) Battery Low Trip Setting (x) Protection against short circuit.(xi) Temperature rise test

The supplier should produce all the relevant original IEC/IS copies of latest amendments/revisions purchased by them for reference by AP Transco representative during acceptance tests.

7.0 INSPECTION:

7.1 The inspection may be carried out by the purchaser at any stage of manufacture. The successful Bidder shall grant free access to the purchaser's representative at a reasonable time when the work is in progress. Inspection and acceptance of any equipment under this specification by the purchaser, shall not relieve the Bidder of his obligation of furnishing equipment in accordance with the specification and shall not prevent subsequent rejection if the equipment is found to be defective.

7.2 The Bidder shall keep the purchaser informed in advance, about the manufacturing Programme so that arrangement can be made for inspection.

7.3 The purchaser reserves the right to insist for witnessing the acceptance/routine testing of the bought out item.

7.4 The bidder shall give 30 days advance intimation to enable the purchaser to depute his representative for witnessing the acceptance and routine tests. The inspection charges

would be to the purchaser's account.

7.5 To & Fro Charges and accommodation shall be provided by the supplier at not lesser than APTransco TA rules.

7.6 APTRANSCO Engineer will not inspect incomplete materials, if APTRANSCO Engineer returned without inspections of offered equipment due to non-readiness of material at their works, the financial loss to the APTRANSCO for deputing Engineer representative will be recovered from the concerned supplier.

7.7 In case of any discrepancy, APTRANSCO has the right to re test the Finished Goods (Type Test and Acceptance Test) at the Cost of Bidder.

8.0 QUALITY ASSURANCE PLAN:

8.1 The bidder shall invariably furnish the following information along with his offer failing which the offer shall be liable for rejection. Information shall be separately given for individual type of equipment offered.

i. The Structure of organization

ii. The duties and responsibilities assigned to staff ensuring quality of work

iii. The system of purchasing, taking delivery and verification of materials

iv. The system for ensuring quality of workmanship

v. The quality assurance arrangements shall confirm to the relevant requirement of ISO-9001 or ISO 9002 as applicable.

vi. Statement giving list of important raw materials, names of sub-supplies for the raw materials, list of standards according to which the raw material are tested, list of tests normally carried out on raw material in the presence of suppliers representative, copies of test certificates.

vii. Information and copies of test certificates as on (i) above in respect of bought out items.

viii. List of manufacturing facilities available

ix. Level of automation achieved and list of areas where manual processing exists.

x. List of areas in manufacturing process, where stage inspections are normally carried out for quality control and details of such tests and inspections.

xi. List of testing equipment available with the bidder for final testing of equipment specified and test plant limitation, if any vis-a-vis the type. Special acceptance and routine tests specified in the relevant standards. These limitations shall be very clearly brought out in "Schedule of Deviations" from the specified test requirements.

8.2 The Contractor shall within 30 days of placement of order, submit the

following information to the purchaser.

i) List of the raw material as well as bought out accessories and the names of sub-suppliers selected from those furnished along with the offer.

ii) Type test certificates of the raw material and bought out accessories if required by the purchaser.

iii) Quality Assurance Plan (QAP) with hold points for purchasers inspection. QAP and purchasers hold points shall be discussed between the purchaser and contractor before the QAP is finalised.

The contractor shall submit the routine test certificates of bought out accessories and central excise asses for raw material at the time of routine testing if required by the purchaser and ensure that the quality assurance requirements of specification are followed by the sub-contractor.

8.3 **The Quality Assurance Programme shall give a description of the Quality System and Quality Plans with the following details.**

i. Quality System:

- The Structure of the Organisation
- The duties and responsibilities assigned to staff ensuring quality of work.
- The system of purchasing, taking delivery of verification of materials.
- The system of ensuring of quality workmanship.
- The system of control of documentation.
- The system of retention of records.
- The arrangement of contractor internal auditing.
- A list of Administrator and work procedures required to achieve contractors quality requirements. These procedures shall be made readily available to the purchaser for inspection on request.

ii) Quality Plans:

- An outline of the proposed work and program sequence.
- The structure of contractors organizations for the contract.
- The duties and responsibilities ensuring quality of work
- Hold and notification points.
- Submission of engineering documents required by this specification.
- The Inspection of the materials and components on request.
- Reference to contractors work procedures appropriate to each activity.
- Inspection during fabrication/ construction.
- Final inspection and test.

8.4 Training of Personnel:

The contractor shall provide necessary facilities for training personnel at their works/principals works relating to testing , operation and maintenance for four personnel free of cost. However, travel and incidental charges of the personnel will be borne by the purchaser.

9.0 DOCUMENTATION:

9.1 All drawings shall conform to International Standards Organization (ISO) 'A' series of drawing sheet/Indian Standards Specification. All drawings shall be in ink and suitable for micro filming. All dimensions and data shall be in S.I. Units.

9.2 List of Drawings and Documents:

The Bidder shall furnish four sets of the following along with his bid.

a) General outline and assembly drawings.

b) Graphics showing the performance of material.

c) General constructional features.

d) Type test reports in case the materials has been typed tested.

e) Test reports literature, pamphlets of the bought out items & raw materials.

9.3 The successful Bidder shall within 2 weeks of placement of order submit four sets of final version of all the above drawings for Purchaser's approval. The Purchaser shall communicate his comments/approval on the drawings to the Supplier within two weeks of receipt of drawings. The supplier shall, if necessary, modify the drawings and resubmit four copies of the modified drawings for Purchaser's approval .The purchaser shall communicate approval with in two weeks of receipt of modified drawings. After receipt of Purchaser's approval, the Supplier shall, within three weeks, submit 20 prints and two good quality re-producibles of the approved drawings for Purchaser's use.

9.4 Six sets of the type test reports, duly approved by the purchaser, shall be submitted by the Bidder for distribution before commencement of supply. Adequate copies of acceptance and routine test certificates, duly approved by the purchaser, shall accompany the dispatch consignment.

9.5 The manufacturing of the equipment shall be strictly in accordance with the approved drawings and no deviation shall be permitted without the written approval of the Purchaser. All manufacturing and fabrication work in connection with the equipment prior to the approval of the drawing shall be at the Bidder’s risk.

9.6 The Bidder shall also furnish twenty copies of bound manuals covering erection, commissioning, operation and maintenance instructions in English Language and all relevant information and drawings pertaining to the main equipment as well as auxiliary devices. Marked erection drawings shall identify the component parts of the equipment as shipped to enable Purchaser to carryout erection with his own personnel. Each manual shall also contain one set of all the approved drawings, type test reports as well as acceptance reports of the corresponding consignment dispatched.

9.7 Approval of drawings by the Purchaser shall not relieve the supplier of any of his

responsibility and liability for ensuring correctness and correct interpretation of the drawings for meeting the requirements of the latest revision of the applicable standards, rules and codes of practices.

9.8 Raw Materials and bought out Items should be approved by APTRANSCO for utilization in manufacture of Batteries.

The equipment shall conform in all respects to high standards of engineering, design, workmanship and latest revisions of relevant standards at the time of supply and Purchaser shall have the power to reject any material which, in his judgment, is not in full accordance therewith.

10.0 PACKING, FORWARDING AND UNLOADING:

The equipment shall be packed in suitable crates so as to withstand handling during transit. The supplier shall be responsible for any damage to the equipment during transit, due to improper and inadequate packing and handling. The easily damageable materials shall be carefully packed and marked with the appropriate caution symbols. Wherever necessary, proper arrangement for lifting, such as lifting hooks etc., shall be provided. Any material found short inside the packing cases shall be supplied by the Supplier without any extra cost.

Each consignment shall be accompanied by a detailed packing list containing the following information:

a) Name of the consignee.
b) Details of consignment.
c) Destination.
d) Total weight of consignment.
e) Handling and unpacking instructions.
f) Bill of material indicating contents of each package and spare material.

The Supplier shall ensure that the packing list and bill of material are approved by the Purchaser before dispatch.

The Supplier shall arrange for unloading the material / equipment at his cost.

11.0 QUANTITY AND DELIVERY REQUIREMENTS:

i. This is set out in the proforma given in Price Schedule and the prices shall be quoted accordingly.

ii. The scope of supply shall include supply of 25% extra-quantity of bolts, nuts, washers, split pins, cotter pins and such other small loose items free of cost.

12.0 SUPERVISION SERVICES:

The supplier shall provide services of qualified personnel for the supervision of erection, testing at site and commissioning the Battery chargers supplied at free of cost.

13.0 GUARANTEED TECHNICAL PARTICULARS:

The Guaranteed Technical particulars for the equipment being supplied shall be provided

with the Bid as specified in this technical specification at Annexure-I**.** The Bids received without guaranteed technical particulars shall be treated as Non-responsive.

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A N N E X U R E – 1

GUARANTEED TECHNICAL PARTICULARS

FOR BATTERY CHARGING EQUIPMENT SUITABLE FOR VRLA AND TUBULAR BATTERY

S.No.	Description	220V 200AH
1	Type of rectifier	
2)	No. of units	
3)	Manufacturer's type designation	
4)	A.C. supply	
	a) Voltage (Volts)	
	b) Current (Amps)	
	c) No. of phases	
	d) Frequency, cycles/Sec	
	e) Power factor	
5)	Percentage taps provided on transformer	
6)	Rated D.C. output for	
	a) Boost Charger (KW)	
	b) Float Charger (KW)	
7)	Rated D.C. output voltage for	
a)	Boost Charger (Volts)	
b)	Float Charger (Volts)	
8)	Rated D.C. output current for	
	a) Boost Charger (Amps)	
	b) Float Charger (Amps)	
9)	Rate of D.C. voltage control for	
	a) Boost Charger (Volts)	
	b) Float Charger (Volts)	
10)	D.C. output voltage regulation from no load to full load volts	
11)	Maximum ripple content %	

S.No.	Description	220V 200AH
12)	Maximum permissible temperature rise over an ambient temperature of 50 Deg.C	
13)	Overall efficiency	
14)	Unit dimensions: Width x Height x Depth mm	
15)	Unit shipping weight Kgs	
16)	Load limiting feature (Please attach characteristics showing variation of voltage with increase in load)	
17)	Recommended settings for thermostat a) Operation temperature Deg. C b) Reset temperature Deg. C	
18)	List of major accessories provided	
19)	Recommended spares	

TECHNICAL SPECIFICATION OF DCDB PANEL for 200AH Charger

1.0 SCOPE:

1.1 This Specification provides for design, engineering, manufacture, assembly, shop testing,inspection and testing before dispatch, packing, forwarding and delivery at Site of DCDB Panels for 200AH charger.

1.2 It is not the intent to specify completely herein all details of the design and construction of equipments. However, the equipment shall conform in all respects to high standards of engineering, design and workmanship and shall be capable of performing continuous commercial operation acceptable to the Purchaser, who will interpret in a manner the meanings of drawings and specifications and shall have the power to reject any material which in his judgment is not in accordance therewith. The equipment offered shall be complete with all components necessary for its effective and trouble -free operation. Such components shall be deemed to be within the scope of Bidder's supply, irrespective of whether these are specifically brought out in this specification and/or the commercial order or not.

2.0 STANDARDS:

All equipment provided under this specification shall in general conform to the latest revision/amendment etc. of the following:

Sl.No.	Standard	Title
1.	IS 3231 IEC 255 Part I to III BS 142	Electrical relays for power system protection
2.	IS 1248 IS 2419 BS 89	Indicating instruments
3.	IS-13010,IS 722(Part Ito IX) IEC 521 BS 3T	Energy meters
4.	IS 4237	General requirement of switchgear and control gear for voltages not exceeding 1 **kV**
5.	IS 375	Marking arrangement for switch gear, busbars, main connection and auxiliary wiring
6.	IS 8686 IEC-255Part V&VI IEC 801-4	Specification for static protective relays and tests.
7.	IEC 337 IEC 337-1 IS 6875	Control switches (LV switching devices for control and auxiliary circuits
8.	IS 2516	MCCB sand load break switches having a break capacity of 30 **kV**A rms at 415 volts
9.	IS 4047	Fuse switching units
10.	IS 2208	HRC fuse links
11.	IS 2705 IEC-185	Current Transformers
12.	IS 3156 IEC-186	Potential Transformers
13.	IS 8623	Factory built assemblies of switch gear and control gear for voltages upto and including 1000V AC and 1200V DC.
14.	IEC 439	Low voltage switch gear and control

		gear assemblies
15.	IS 4237	General requirement of switch- gear and control gear for voltages not exceeding 1000V
16.	IS 2147	Degree of protection provided by enclosures for low voltage switch gear and Control gear
17.	IS 5082	Aluminum alloy bars
18.	IS 1897	Copper strips for electrical purpose
19.	IS 3618 IS 6005	Code of practice for phosphate treatment of iron and steel
20.	IS 4064	Air break switches, air break dis-connectors, air break disconnectors and fuse combination units for voltages not exceeding 1000V AC or 1200V DC
21.	IS 9224 Part II	HRC catridge fuses
22.	IS 5578	Guide for marking of insulated conductors.
23.	IS 2032	Graphical symbols Ref- abbreviations

Equipment meeting with the requirements of other authoritative Standards, which ensure equal or better performance than the standards mentioned above, shall also be considered. When the equipment offered by the Bidder conforms to other standards, salient points of difference between standards adopted and the standards specified in this specification shall be clearly brought out in the relevant schedule. Four copies of such standards with authentic translation in English shall be furnished along with the offer.

3.0 CLIMATIC CONDITIONS:

3.1 The equipment to be supplied under this specification shall be suitable for satisfactory operation under the following climatic conditions.

i)	Location	Andhra Pradesh
ii)	Max. ambient air temp. (deg.C)	50
iii)	Min ambient air temp. (deg.C)	0
iv)	Average daily ambient air temperature(deg.C)	35

v)	Maximum Relative humidity (%)	80
vi)	Average annual rainfall (mm)	925
vii)	Isoceraunic Level (Days/Annum)	50
viii)	Max. wind pressure (Kg/sq.m.)	200
ix)	Seismic level (Horizontal Acceleration)	0.3g
x)	Max. altitude above mean sea level (Metres)	1000
xi)	Average No. of thunder storms days/ annum.	40
xii)	Average No. of rainy days/ annum.	90
xii)	Average No. of Months Tropical monsoon conditions/ annum.	

4.0 **PRINCIPAL PARAMETERS:** The technical details of various equipments such as power transformers, circuit breakers and instrument transformers that will be associated with the control and relay panels in APTRANSCO are to be known by the bidder.

5.0 **D.C. DISTRIBUTION BOARD :**

5.1 The D.C. Distribution boards shall be provided with 2Nos. bus-bars made out of electrolytic copper of adequate capacity duly tinned supported by bus-bar supporters and are placed at sufficient distance instead of side by side to avoid accidental short circuit.

These busbars will be connected to 220V D.C. 2 wire system consisting of 220V, 200AH batteries or battery Charger rated for 220V D.C., 25A output.

The D.C. distribution boards required under this specification are to feed control circuits, and other auxiliary DC equipment of the Substations. DCDB shall be Free Standing, Floor Mounted Type Sheet Steel of CRCA, Non- Compartmental Type.

5.2 The D.C. distribution boards shall be of rugged construction, designed for working satisfactorily under the severest operating and climatic conditions and shall conform to the relevant ISS of latest issue.

5.3 The D.C. distribution boards shall be complete with all accessories and fittings necessary for the efficient and satisfactory operation whether they are specifically mentioned or not in the specification. The D.C. distribution boards shall be provided with the necessary reputed make switches, fuses, or DC Circuit breakers in the incoming and outgoing circuits and the equipment offered shall be complete in itself.

5.4 The D.C. distribution board shall comprise the following components for 220V, 25A,200AH.

The D.C distribution board shall consist of provision for two 220 V DC input sources with 2 Nos Copper bus bar per each input source and DC Bus coupler arrangement. The outgoing feeders of DCDB shall be equally distributed between two sources of DC supply

220V, 200AH

a. Double pole MCB for incoming DC supply	3 Nos.(100 A)
b. Double pole MCB for bus coupler	*1 Nos*
c. Double pole 20 A MCB Outgoing feeders	24 Nos
d. Double pole 10 A MCB Outgoing feeders	48 Nos

5.5 All the D.C. MCB/MCCBs shall be mounted on the front of the panel. The operating handle shall protrude out of the panel.

5.6 The D.C. distribution Board shall be of simplex type, sheet steel cubicle, indoor type, floor mounted and self supporting. It shall be totally enclosed, completely dust tight, weather and vermin proof. Ingress protection of DCDB cabinet shall be IP 42

5.7 The D.C. distribution board shall be made out of suitable angles and M.S sheet of 16 SWG for the front, 18 SWG for the back, top and sides, and 14 SWG for the bottom. The D.C. distribution board height shall be 1800mm,depth shall be 600 mm and suitable width to accommodate all components i.e., adequate clearance/space should be provided between item to item to facilitate easy handling at the time of attending on repairs. The D.C. distribution Board shall be provided with hinged type rear doors.

5.8 The internal wiring of the D.C. distribution board shall be carried out with PVC insulated FRLSH, 1100V grade stranded copper conductor. The control wiring shall be carried out with 2.5 Sq.mm copper conductors. All wiring for external connections shall be brought to 1100V grade single piece molded terminal blocks with barriers. Suitable terminal blocks with stud type terminals shall be provided. The cables used shall have ISI marking on them. Extra 20% spare terminal blocks shall be provided.

5.9 The DC distribution board shall be provided with indicating LED lamps to indicate DC supply ON position, for all circuits. Provision shall be made for writing the Name of the Circuit and necessary stickers shall be provided.

5.10 Digital multi-function meter with RS 485 port shall be provided for measuring DC voltage and current for both DCDB incomers.

5.11 The DC distribution boards shall be complete in all respects with all internal wiring incoming and outgoing terminals, labels, earthing connections and required cable glands.

5.12 All the final drawings and BOM of panels shall be approved by Design wing of APTRANSCO.

5.13 The DCDB shall be offered complete with all parts that are necessary or usual for their efficient operation. Such parts shall be deemed to be within the scope of this specification, whether specifically mentioned or not.

5.14 All cubicles shall undergo tank cleaning process according to latest IS in vogue followed by powdered coating process. The painting shall be uniform appearance and semi glossy finish (visually) without blistering. There shall not be any peeling at intersecting points or joints.

5.15 Miscellaneous Accessories:

Plug point- 230 Volts, single phase, 50 Hz AC plug point shall be provided in the interior of each cubicle with "ON-OFF" Switch for connection of hand lamps.

Interior lighting: Each panel shall be provided with a LED BULB rated for 230 volts, single phase, 50 HZ supply for the interior illumination of the panel during maintenance. The fittings shall be complete with switch fuse unit and switching of the lighting shall be controlled by the respective panel door switch. **Toggle Switch shall be provided in addition to the Door Limiting Switch.**

ON/OFF Switch shall be provided in addition to Thermostat in the Space Heater Circuit.

Exclusive Fuse Protection shall be provided in the PS Circuit (6A HRC Fuse & Link), Panel Lighting Circuit (2A HRC Fuse & Link) and in the Space Heater Circuit (2A HRC Fuse & Link) in addition to MCB in the Incoming AC Circuit.

Supporting steel: All necessary foundation anchor bolts and other parts of supporting and fastenings, of the panels shall be supplied by the contractor.

5.16 Earthing:

All panels shall be equipped with an earth bus securely screwed. Location of earth bus shall ensure no radiation interference for earth systems under various switching conditions of isolators and breakers. The material and the sizes of the bus bar shall be atleast 25 x 3 mm copper flat.

When several panels are mounted adjoining each other, the earth bus shall be made continuous and necessary connectors and clamps for this purpose shall be included in the scope of supply of contractor. Provision shall be made for extending the earth bus bars to future adjoining panels on either side.

Provision shall be made on each earth bus bars of the panels for connecting Purchaser's earthing grid. Necessary terminal clamps and connectors for this purpose shall be included in the scope of supply of contractor. The wire or screens should be clearly bonded and earthed at the gland plate.

All metallic cases of relays, instruments and other panel mounted equipment shall be connected to the earth bus preferably by independent copper wires of size not less than 2.5 sq.mm. The colour code of earthing wires shall be green. Earthing wire shall be connected on terminals with suitable clamp connectors and soldering shall not be permitted.

Looping of earth connections which would result in loss of earth connection to other devices where the loop is broken shall not be permitted. However, looping of earth connections between equipment to provide alternative paths to earth bus shall be provided.

6.0 TESTS

The bidder shall carryout routine tests on all associated equipment as per relevant Indian Standards/IEC Standards. The bidder shall submit along with the bid type test reports for protective relays, instruments, switches and other devices of the type and class being offered to assess the suitability of the equipment. Type tests must have been conducted not earlier than Ten years as on the date of bid opening. The bids received without type test reports will be treated as non-responsive.

ALL panels shall be subjected to the following tests:

a. Verification of degree of protection as per IS:2147 (TypeTest)

b. High voltage test (2 kV for 1 minute) or as per relevant IS.

c. Insulation resistance test

d. Electrical control, interlock and sequential operation test

e. Verification of wiring and general arrangement as per approved schematic/G.A. diagrams.

7.0 Inspection:

7.1 i All routine tests shall be conducted at the time of inspection and inspection shall be made at the place of manufacturers unless otherwise specifically agreed upon by the manufacturer and purchaser at the time of purchase. The manufacturer shall offer the inspector representing the purchaser all reasonable facilities without charges to satisfy him that the panels/materials are manufactured/supplied as per the specification/final approved drawings.

ii. The supplier shall give 15 days (for local supplier) advance intimation to enable the purchaser to depute his representative for witnessing acceptance and routine tests.

iii. The panels shall be manufactured and offered for inspection as per the final approved General Arrangement Drawing and schematic drawing which will be binding both on supplier and purchaser.

iv. The panels shall not be dispatched from its point of manufacture unless the same are satisfactorily inspected and tested.

v. The acceptance of any material shall in no way release the supplier of his responsibility for meeting all the requirements of the specification and shall not prevent subsequent rejection of such material later found to be defective.

vi. The To and fro travelling expenses and accommodation facility for inspection officers shall be borne by the supplier of material / Bidder.

8.0. Quality Assurance Plan:

8.1 The bidder shall invariably furnish the following information along with his offer failing which the offer shall be liable for rejection. Information shall be separately given for individual type of equipment offered.

i. The Structure of organization

ii. The duties and responsibilities assigned to staff ensuring quality of work

iii. The system of purchasing, taking delivery and verification of materials

iv. The system for ensuring quality of workmanship

v. The quality assurance arrangements shall conform to the relevant requirement of ISO-9001 and ISO 9002 as applicable.

vi. Statement giving list of important raw materials, names of sub-supplies for the raw materials, list of standards according to which the raw material are tested, list of tests normally carried out on raw material in the presence of suppliers representative, copies of test certificates.

vii. Information and copies of test certificates as on (i) above in respect of bought out items.

viii. List of manufacturing facilities available

ix. Level of automation achieved and list of areas where manual processing exists.

x. List of areas in manufacturing process, where stage inspections are normally carried out for quality control and details of such tests and inspections.

xi. List of testing equipment available with the bidder for final testing of equipment specified and test plant limitation, if any vis-a-vis the type. Special acceptance and routine tests specified in the relevant standards. These limitations shall be very clearly brought out in "Schedule of Deviations" from the specified test requirements.

8.2 The Contractor shall within 30 days of placement of order, submit the following information to the purchaser.

i. List of the raw material as well as bought out accessories and the names of sub-suppliers selected from those furnished along with the offer.

ii. Type test certificates of the raw material and bought out accessories if required by the purchaser.

iii. Quality Assurance Plan (QAP) with hold points for purchasers inspection. QAP and purchasers hold points shall be discussed between the purchaser and contractor before the QAP is finalised.

The contractor shall submit the routine test certificates of bought out accessories and central excise asses for raw material at the time of routine testing if required by the purchaser and ensure that the quality assurance requirements of specification are followed by the sub-contractor.

8.3 The Quality Assurance Programme shall give a description of the Quality System and Quality Plans with the following details.

A. Quality System:

i) The Structure of the Organization
ii) The duties and responsibilities assigned to staff ensuring quality of work.
iii) The system of purchasing, taking delivery of verification of materials.
iv) The system of ensuring of quality workmanship.
v) The system of control of documentation.
vi) The system of retention of records.
vii) The arrangement of contractor internal auditing.
viii) A list of Administrator and work procedures required to achieve contractors quality requirements. These procedures shall be made readily available to the purchaser for inspection on request.

B. Quality Plans:

a. An outline of the proposed work and program sequence.
b. The structure of contractors organizations for the contract.
c. The duties and responsibilities ensuring quality of work
d. Hold and notification points.
e. Submission of engineering documents required by this specification.
f. The Inspection of the materials and components on request.
g. Reference to contractors work procedures appropriate to each activity.

h. Inspection during fabrication/construction.

i. Final inspection and test.

9.0 Documentation:

9.1 The bidder shall furnish the dimensioned drawings along with the bid indicating the details like components, make, quantity, ranges of meters, relays etc., in case the bidder quotes different makes of relays shall furnish copies of wiring scheme and literature of all relays.

10.0 Packing and Forwarding:

10.1 Name plate (giving details of equipment) in all equipments (i.e. relays, breaker switches, indicating lamps, push buttons etc.) mounted on the panel shall be provided. Name plates of suitable size on metal/PVC acrylic/thick plastic sheets shall only be provided. Stickers in no case will be accepted. Large and bold name plates shall be provided for circuit/feeder designation. Name plates for AC fuses, DC fuses, resistors and other equipment mounted inside the panel shall be provided clearly indicating the purpose for which they are meant for.

Name plates of suitable size shall be provided on the top of the panel clearly indicating manufacturers name, purchase order number and date, year of manufacturer, serial number of the panel and type of panels. The panels shall be packed in suitable sized strong and weather resistant wooden cases / crates.

10.2 Suitable cushioning, protective padding or spacer shall be provided to prevent damage to or deformation of the components during transit and handling.

10.3 All packing cases shall be marked legibly and correctly with the suppliers name and consignee's address so as to ensure their safe arrival at their destination and to avoid possibility of goods being lost or wrongly dispatched on account of faulty or illegible marking.

11.0 Coordination:

11.1 The bidder shall coordinate in all respects with the manufacturers of associated equipment and shall freely exchange all relevant information to ensure satisfactory and economic overall design. No extra charge shall be payable for these services. The bidder is responsible for the performance and service support for all the bought-out items he has used in the panel.

11.2 Erection of equipment called for in this specification will be undertaken by the purchaser. The bidder shall furnish full detailed instructions, complete with drawings and leaflets to enable satisfactory erection and commissioning of the plant by the departmental staff. Such instructions should reach the purchaser sufficiently in advance of the receipt of the equipment in order that all the preliminary arrangement for the erection of the equipment can be made by the department in advance so that there will be no delay in regard to erection as soon as the equipment are received.

It is obligatory on the part of the bidder to make available services of their testing and commissioning engineers (with necessary testing equipment) to assist the APTRANSCO staff in commissioning their equipment as and when their services are requisitioned.

12.0 Tropical Treatment:

12.1 Under the climatic conditions prevalent at site, the equipment furnished under the specification will be subjected to operation under ambient temperatures of 50 deg. C and very high relative humidity. All equipment shall therefore, be suitably designed and treated for normal life and satisfactory operation under the hot and humid climatic conditions prevalent at site and shall be dust and vermin proof. All parts and surfaces which are subject to corrosion, shall be made of such material and shall be provided with such protective finishes as would protect the equipment installed from any injurious effects of excessive humidity. All electrical auxiliary equipment shall be specifically treated for tropical conditions, and the materials and methods for this treatment shall be approved in advance by the Purchaser.

13.0 Foundations and Fixtures:

Foundations will be prepared by the Purchaser in accordance with the basic designed data to be supplied by the contractor. All fixtures necessary for proper erection of the embedded parts shall be supplied by the supplier if required.

DCDB PANELS:

TECHNICAL PARTICULARS

A. Instruments (Digital Voltmeter and Ammeter)

a. Make
b. Type
c. Applicable standards
d. Range
e. Shape
f. Size
g. Scale Length (mm)
h. Finish
i. Temperature at which calibrated

B. Bus bars and Insulators

1. Material of bus bars and applicable standards

2. Cross sectional area of
   bus bars in Sq. mm
   i) For horizontal bus bar
   ii) For vertical bus bar
3. Type of bus bar insulation
   (Bare/Painted/Epoxy insulated
   sleeved)
4. Minimum clearance (in mm)
   a) Phase to phase
   b) phase to earth
5. Continuous current rating
   At 50 Deg. Ambient temperature
6. Short time current rating
   for 1 sec.
7. Material of the support
   Insulator

C. MCCBs/MCBs

1. Make and type
2. Rated current at design
3. Continuous current at
   site ambient temperature
4. Rated symmetrical
   breaking current
5. Rated making current
6. Short time current rating
7. Earth Bus
   1) Material
   2) Cross sectional area (Sq. mm)

D. Panels

1. Material and thickness of
   a) Front sheet (mm)
   b) Back sheet (mm)
   c) Barrier, sheet between

units (mm)

d) End covers (mm)

2. Dimension of Panels (L x H x D)
3. Symmetrical short circuit withstand current of switchgear cubicle in KA
4. Degree of protection as Per IS 2147
5. Width of cable alley
6. Total weight of the panel including all the enclosures
8. Painting (powder Coated)
   a) Colour
   b) Finish
   c) No. of coats